

Master of Science in Industrial and Applied Mathematics  
Specialization in Data Science

# **Statistical Analysis of Stochastically Quantized Random Features in SGD for Memory-Constrained Kernel Learning**

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## Abstract

Kernel methods are widely used in machine learning: they implicitly embed the data into a reproducing kernel Hilbert space (RKHS), so that linear estimators in the feature space yield non-linear predictors in the input space, with well-understood statistical guarantees. Their drawback is cost, in both time and space: forming and factorizing the kernel matrix takes  $O(n^2)$  memory and  $O(n^3)$  time in the sample size  $n$ . Random Fourier features (RFFs) approximate the kernel by the inner product of an explicit  $M$ -dimensional feature map, which the learning algorithm can use in place of kernel evaluations, and lower this to  $O(nM)$  memory and  $O(nM^2)$  time, with  $M \ll n$ . On a memory-constrained system, however, storing  $nM$  floating-point numbers is still often out of reach.

We analyze the loss of statistical accuracy when those features are stochastically rounded to one bit per coordinate. Building on the analysis of mini-batch stochastic gradient descent (SGD) with random features of [4], we study least-squares regression solved by mini-batch SGD on one-bit RFFs, so that each coordinate is stored on a single bit instead of the 32 of single precision. We bound the excess risk  $\mathcal{E}(f_T^q) - \mathcal{E}(f_{\mathcal{H}})$  of the resulting estimator, where  $f_{\mathcal{H}}$  minimizes the expected square risk over the RKHS  $\mathcal{H}$  associated with the target kernel. Rounding perturbs the covariance operator by a diagonal term of order  $1/M$ , which produces a bias between the quantized and the full-precision trajectories and an extra gradient variance. Both are controlled, and we obtain an excess risk of order  $\tilde{O}(1/\sqrt{n})$  with  $M = \tilde{O}(\sqrt{n})$  features, the same number needed in full precision; this rate is optimal in a minimax sense [3]. The bound rests on a spectral condition on the mini-batch operators, which we verify in the population case and leave open in general.

We then test this on the CASP and SUPERCONDUCT datasets, with the Gaussian and Matérn kernels. The statistical price of quantization is paid only when features are scarce: at  $M = 10$  the test error exceeds the full-precision one by about ten per cent, and the gap has closed by  $M \simeq 10^3$ , well inside the regime the theory prescribes. At  $M = 10^4$  the stored feature matrix shrinks from 320 MB to 10 MB.

## Résumé

Les méthodes à noyau sont largement utilisées en apprentissage automatique : elles plongent implicitement les données dans un espace de Hilbert à noyau reproduisant (RKHS), de sorte que des estimateurs linéaires dans l'espace des caractéristiques fournissent des prédicteurs non linéaires dans l'espace d'entrée, avec des garanties statistiques bien comprises. Leur inconvénient est le coût, en temps comme en espace : construire et factoriser la matrice de noyau demande  $O(n^2)$  en mémoire et  $O(n^3)$  en temps, où  $n$  est la taille de l'échantillon. Les caractéristiques de Fourier aléatoires (RFFs) approchent le noyau par le produit scalaire d'une application de caractéristiques explicite de dimension  $M$ , que l'algorithme d'apprentissage peut utiliser à la place des évaluations du noyau, et ramènent ce coût à  $O(nM)$  en mémoire et  $O(nM^2)$  en temps, avec  $M \ll n$ . Sur un système à mémoire contrainte, cependant, stocker  $nM$  nombres à virgule flottante reste souvent hors de portée.

Nous analysons la perte de précision statistique lorsque ces caractéristiques sont arrondies stochastiquement à un bit par coordonnée. En nous appuyant sur l'analyse de la descente de gradient stochastique (SGD) par mini-lots avec caractéristiques aléatoires de [4], nous étudions la régression aux moindres carrés résolue par SGD par mini-lots sur des RFFs à un bit, chaque coordonnée étant ainsi stockée sur un seul bit au lieu des 32 de la simple précision. Nous bornons l'excès de risque  $\mathcal{E}(\hat{f}_T) - \mathcal{E}(f_{\mathcal{H}})$  de l'estimateur obtenu, où  $f_{\mathcal{H}}$  minimise le risque quadratique espéré sur le RKHS  $\mathcal{H}$  associé au noyau cible. L'arrondi perturbe l'opérateur de covariance par un terme diagonal d'ordre  $1/M$ , ce qui produit un biais entre les trajectoires quantifiée et en pleine précision ainsi qu'une variance de gradient supplémentaire. Ces deux termes sont contrôlés, et nous obtenons un excès de risque d'ordre  $\tilde{O}(1/\sqrt{n})$  avec  $M = \tilde{O}(\sqrt{n})$  caractéristiques, soit le même nombre qu'en pleine précision ; cette vitesse est optimale au sens minimax [3]. La borne repose sur une condition spectral portant sur les opérateurs de mini-lots, que nous vérifions dans le cas de la population et laissons ouverte en général.

Nous testons ensuite ces résultats sur les jeux de données CASP et SUPERCONDUCT, avec les noyaux gaussien et de Matérn. Le prix statistique de la quantification n'est payé que lorsque les caractéristiques sont peu nombreuses : à  $M = 10$ , l'erreur de test dépasse celle de la pleine précision d'environ dix pour cent, et l'écart est résorbé dès  $M \simeq 10^3$ , bien à l'intérieur du régime prescrit par la théorie. À  $M = 10^4$ , la matrice de caractéristiques stockée passe de 320 Mo à 10 Mo.

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# Introduction

This thesis studies kernel learning under a memory constraint. Given  $n$  input–output pairs, the goal is to learn a predictor that generalizes to new data. Kernel methods are well suited to this setting: they represent nonlinear predictors through linear methods in a reproducing kernel Hilbert space. Their main drawback is computational. Standard kernel algorithms rely on the  $n \times n$  kernel matrix, which requires  $\mathcal{O}(n^2)$  memory and typically  $\mathcal{O}(n^3)$  time for factorization [6]. This quickly becomes impractical for large datasets.

Random Fourier features (RFFs) [12] avoid constructing the kernel matrix explicitly. They map each input to an  $M$ -dimensional random feature vector whose inner products approximate the target kernel in expectation. This reduces the memory cost to  $\mathcal{O}(nM)$ . For the learning setting considered here,  $M = \tilde{\mathcal{O}}(\sqrt{n})$  features are sufficient to retain the statistical rate of the full kernel method [4].

Even then, storing the feature matrix can remain expensive. A dataset represented by  $nM$  single-precision values requires  $32nM$  bits. This can be too large for memory-limited devices, even when  $M \ll n$ . We therefore consider one-bit representations of the random features.

There are two natural one-bit schemes. Deterministic quantization stores only the sign of each feature. It is simple and performs well empirically, but it changes the kernel being approximated: the resulting inner products are biased with respect to the target kernel. Stochastic quantization instead rounds each coordinate to one of two values with probabilities chosen so that the quantized coordinate remains unbiased. It preserves the target feature map in expectation, but introduces additional noise during SGD.

This thesis studies stochastic one-bit quantization. The point is to isolate the statistical effect of quantization from the separate question of which kernel or RKHS is induced by a deterministic binary map. Since quantization is applied to the features used at every SGD iteration, its effect accumulates along the optimization trajectory. We analyze this additional source of error and quantify its impact on the final estimator.

## Summary of contributions

The technical statement of what follows uses notions — reproducing kernel Hilbert space, translation invariant kernel, Bochner’s theorem — introduced in Sections 1.1 and 1.2.

**Position.** For a normalized translation invariant kernel  $\kappa^\Delta$  and the dithered map  $\phi_M(x) = \sqrt{2/M} \cos(\Omega^\top x + \xi)$ , [4] show that mini-batch SGD on  $\phi_M$  attains the minimax rate of [3],

$$\mathbb{E}_J \|f_T - f_{\mathcal{H}}\|_{L^2(\rho_{\mathcal{X}})}^2 = \mathcal{O}(n^{-1/2}), \quad M = \tilde{\mathcal{O}}(\sqrt{n}), \quad \mu T \asymp \sqrt{n},$$

with no explicit penalty term. We investigate what is the effect of storing each coordinate of  $\phi_M$  on one bit. Two schemes are available. Stochastic rounding sends  $\phi_M(x)_j$  to  $\pm R$ ,  $R := \sqrt{2/M}$ , with  $\text{Prob}(\phi_M^q(x, \gamma)_j = \pm R) = \frac{1}{2}(1 \pm \cos(\omega_j^\top x + \xi_j))$ , so that  $\mathbb{E}_\gamma[\phi_M^q(x, \gamma)] = \phi_M(x)$ ; deterministic rounding sends it to  $\phi_M^{\text{det}}(x)_j = M^{-1/2} q(\omega_j^\top x + \xi_j)$ ,  $q = \text{sgn} \circ \cos$ . The two differ in the kernel they reproduce: writing  $Q_h = \frac{2}{\pi h} (-1)^{(h-1)/2}$  for the Fourier coefficients of  $q$ , with  $\sum_{h \text{ odd}} |Q_h|^2 = 1$ , we have:

$$\begin{aligned} \mathbb{E} \langle \phi_M^q(x, \gamma), \phi_M^q(y, \gamma') \rangle &= \kappa^\Delta(x - y), \\ \mathbb{E} \langle \phi_M^{\text{det}}(x), \phi_M^{\text{det}}(y) \rangle &= \sum_{h \text{ odd}} |Q_h|^2 \kappa^\Delta(h(x - y)) =: \kappa_{q,q}^\Delta(x - y) \end{aligned} \quad (1.1)$$

by Proposition 2.5 of [16], where the latter expectation is taken w.r.t.  $\Omega, \xi$ , while the former also w.r.t.  $\gamma$ . We analyze the first case since it is the one that keeps the model fixed, and it is the one we are interested in.

**Result.** The estimator is the recursion of [4] with  $\phi_M$  replaced by  $\phi_M^q$ ,

$$w_{t+1}^q = w_t^q - \frac{\mu}{b} \sum_{i \in B_t} (\langle w_t^q, \phi_M^q(x_i, \gamma_{t,i}) \rangle_{\mathbb{R}^M} - y_i) \phi_M^q(x_i, \gamma_{t,i}), \quad (1.2)$$

and its excess risk is not covered by existing analyses: [20] quantize the same features but bound a kernel-matrix approximation quantity, and [16] work outside training. Rounding perturbs the covariance operator by a diagonal term,

$$C_M^q = C_M + \bar{D}, \quad \bar{D} = \mathbb{E}_\gamma[\eta \eta^\top] \text{ diagonal}, \quad \|\bar{D}\|_{op} \leq \frac{2}{M},$$

and we propagate this perturbation through (1.2) in  $L^2(\rho_{\mathcal{X}})$ , obtaining

$$\mathbb{E}_{J,\gamma} \|f_T^q - f_{\mathcal{H}}\|_{L^2(\rho_{\mathcal{X}})}^2 \lesssim \underbrace{\sigma_{\mathcal{H}}^2 \Xi_T}_{\asymp \mu \log T / b} + \underbrace{\frac{U^2}{M^2} \Theta_T^2}_{\asymp \mu T / M^2} + \mathbb{E}_J \|f_T - f_{\mathcal{H}}\|_{L^2(\rho_{\mathcal{X}})}^2,$$

where  $\Theta_T, \Xi_T$  are spectral sums controlled under an assumption on the mini-batch operators that we verify in the population case. Since  $\mu T \asymp \sqrt{n}$  in every admissible configuration of [4], both added terms are  $\tilde{\mathcal{O}}(n^{-1/2})$  as soon as  $M \gtrsim \sqrt{n}$ : one bit per coordinate preserves the optimal rate with the same number of features as in full precision, the binding requirement being the bias.

**Experiments.** On CASP and SUPERCONDUCT with Gaussian and Matérn kernels, the quantized error exceeds the full-precision one noticeably for  $M \leq 100$  and the gap closes by  $M \simeq 10^3$ . Freezing the dither, which is what actually removes the  $nM$  floats — 320 MB against 10 MB at  $M = 10^4$  — leaves the curves unchanged although it falls outside the analysis. Deterministic quantization is competitive with full precision and sometimes better.

**A first step towards the deterministic estimator.** That last observation is not a weakness of the analysis but the reason for it. Given  $(\Omega, \xi)$  the deterministic map is non-random, so there is no rounding noise to propagate: (1.2) with  $\psi(x, (\omega, \xi)) = q(\omega^\top x + \xi)$ ,  $|\psi| \equiv 1$ , is just the algorithm of [4] for the random feature  $\psi$ . What changes is the model. By (1.1) its kernel is  $\kappa_{q,q}^\Delta$ , and since it is a weighted series of  $k^\Delta$ , as seen in [6], we have that  $\mathcal{H}_{\kappa^\Delta} \subseteq \mathcal{H}_{\kappa_{q,q}^\Delta}$ : hence a rate for it would be something of the form:  $\mathbb{E}\|f_T^{\text{det}} - f_{\mathcal{H}_{q,q}}\|^2 = \mathcal{O}(n^{-1/2})$ , hence about a *different* target function. Therefore, from an analytical perspective, the model we analyze allows us to separate the effect of quantization from the effect of being in a bigger function space. So that the gap observed in Chapter 4 is attributable to the model and not to the bit approximation. The remaining step is then approximation-theoretic and may requires tools different from the ones in our analysis and the one in [4], since the quantization map  $q$  is no longer continuous, so Assumption 2 in [4] fails. A proposal is to substitute it by Lipschitz smoothness as in [16], but we leave the analysis open.

## 1.1 Overview of Statistical Machine Learning

Machine learning, a branch of artificial intelligence, focuses on designing algorithms and statistical models that infer underlying patterns from data to perform tasks without relying on explicit, rule-based instructions.. The core idea behind machine learning is to allow systems to learn from data, identify patterns, and make decisions with minimal human intervention on data not previously seen. Machine learning algorithms can be broadly categorized into three main types: supervised learning, unsupervised learning, and reinforcement learning. We are going to focus on supervised learning, which is the most widely used type of machine learning, where the algorithm is trained on an input-output dataset, meaning that each training example is paired with an output value (often referred to as a label in classification tasks, or a target in regression tasks).

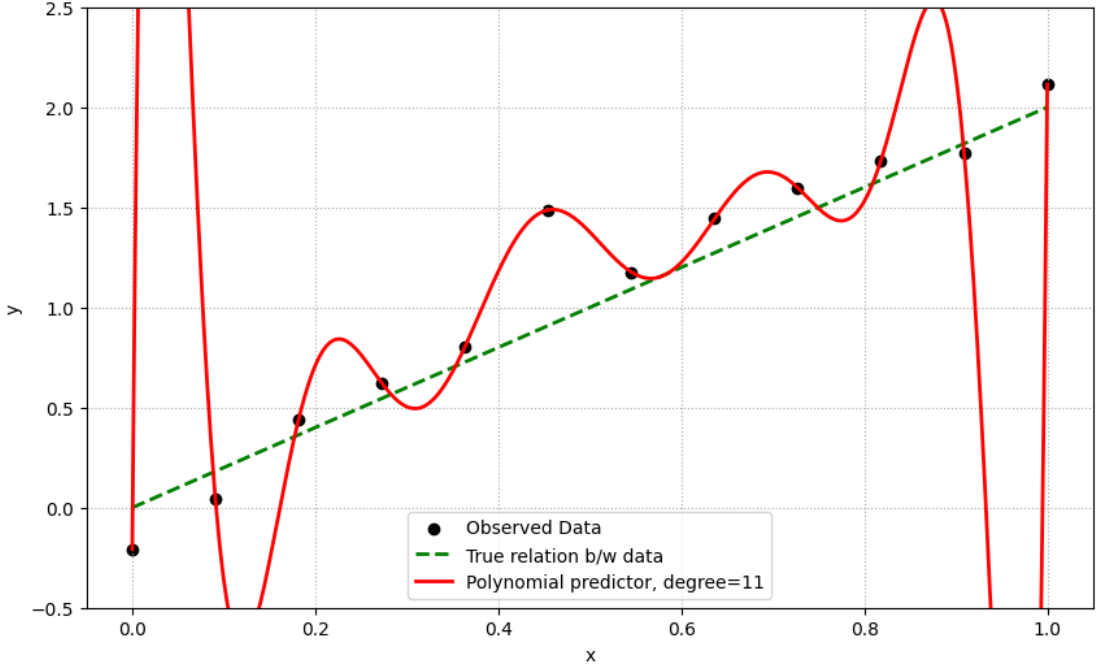


Figure 1.1: Example of a poor function space choice. The complexity of the polynomial model allows it to interpolate the training data, leading to severe oscillations in regions with low sample density.

Assume we have a dataset of  $n$  examples, where each example consists of an input vector  $x_i \in \mathbb{R}^d$  and a corresponding output target  $y_i \in \mathbb{R}$ . The goal of supervised learning is to learn a function  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  that can predict the output  $y$  for new, unseen input vectors  $x$ . To do so, we "train" the model, i.e. we look for the best function in a certain function class, by a risk function that measures the discrepancy between the predicted outputs and the true targets in the training data, i.e. we solve the following optimization problem:

$$\min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n L(f(x_i), y_i) + \lambda R(f). \quad (1.3)$$

Here  $\mathcal{H}$  is a suitable space of functions (predictors),  $L$  is a loss function that quantifies the error of the predictions,  $R$  is a regularization term, i.e. it prevents  $f$  from varying too sharply in order to fit the known data hence only "memorizing" it. This necessity arises because we only have access to a limited number of samples making the model susceptible to spurious oscillations and  $\lambda$  is a regularization parameter that controls the trade-off between fitting the training data and keeping the model simple. One of the most common choices for the function  $L$  is the square loss, defined as  $L(f(x_i), y_i) = (f(x_i) - y_i)^2$ , which is widely used for regression problems and the one we will be using in this work. While a common choice for the regularization term  $R$  is the squared norm

of  $f$  in a suitable function space.

As previously mentioned, choosing the right function space  $\mathcal{H}$  is crucial for the success of the learning algorithm. In the classical ML setting, we observe that if the space is too large, in a sense that we will further clarify in the next sections, the model may overfit the training data, meaning that it will perform well on the training data but poorly on new, unseen data (although it is worth noting that while this poor generalization is typical in the classical statistical machine learning setting, in some modern contexts like overparametrized deep networks, models can interpolate the data and still generalize very well). On the other hand, if the space is too small, the model may underfit the training data, meaning that it will perform poorly on both the training data and new, unseen data. It is also important to work with a function space that allows for efficient optimization, as the training process involves solving an optimization problem, which could be unfeasible if the function space is too complex. In the example given in Figure 1.1 we allow the function space to include highly interpolating polynomials, which leads to a zero training error but once evaluated on new data, the model fails to generalize and performs poorly. While we generally speak of overfitting when a model fits the noise, in this specific case, the wild oscillations on the sides are primarily due to the polynomial nature of the model combined with a scarcity of points, providing a clear example of how a poorly chosen function space fails to capture the underlying phenomenon. We give now more formal definitions.

**Definition 1.1.1.** Let  $\mathcal{X} \subseteq \mathbb{R}^d$  be the input space and  $\mathcal{Y} \subseteq \mathbb{R}$  the output space. Over the product space  $\mathcal{X} \times \mathcal{Y}$  we assume there is a probability distribution function  $\rho$  that generates the data, and we denote by  $\rho_x$  the marginal distribution of  $\rho$  on  $\mathcal{X}$  and with  $\rho(y|x)$  the conditional distribution of  $\rho$  given  $x$ .

We also define  $\mathcal{Z} = (x_i, y_i)_{i=1}^n \subseteq (\mathcal{X} \times \mathcal{Y})^n$  the given dataset where the  $(x_i, y_i)$  are i.i.d. samples drawn accordingly to  $\rho$ .

As previously mentioned, the goal of supervised learning is to learn from a finite number of noisy observations a function  $f : \mathcal{X} \rightarrow \mathcal{Y}$  that can predict the output label  $y$  for new, unseen input vectors  $x$ ; so the next definition comes naturally:

**Definition 1.1.2.** Let  $\mathcal{H}$  a space of function from  $\mathcal{X}$  to  $\mathcal{Y}$ , each of the function  $f$  in  $\mathcal{H}$  is called a predictor.

Now we define the quantity with respect to which we choose the best predictor:

**Definition 1.1.3.** A function  $L : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}$  is called loss function. In the following, we use the mean squared error, defined as:

$$l(y', y) := \|y' - y\|_{\mathbb{R}}^2$$

**Definition 1.1.4.** We call expected risk the function:

$$\mathcal{E}(f) := \mathbb{E}_{\rho}(l(f(x), y)) = \int_{\mathcal{X} \times \mathcal{Y}} l(f(x), y) d\rho(x, y)$$

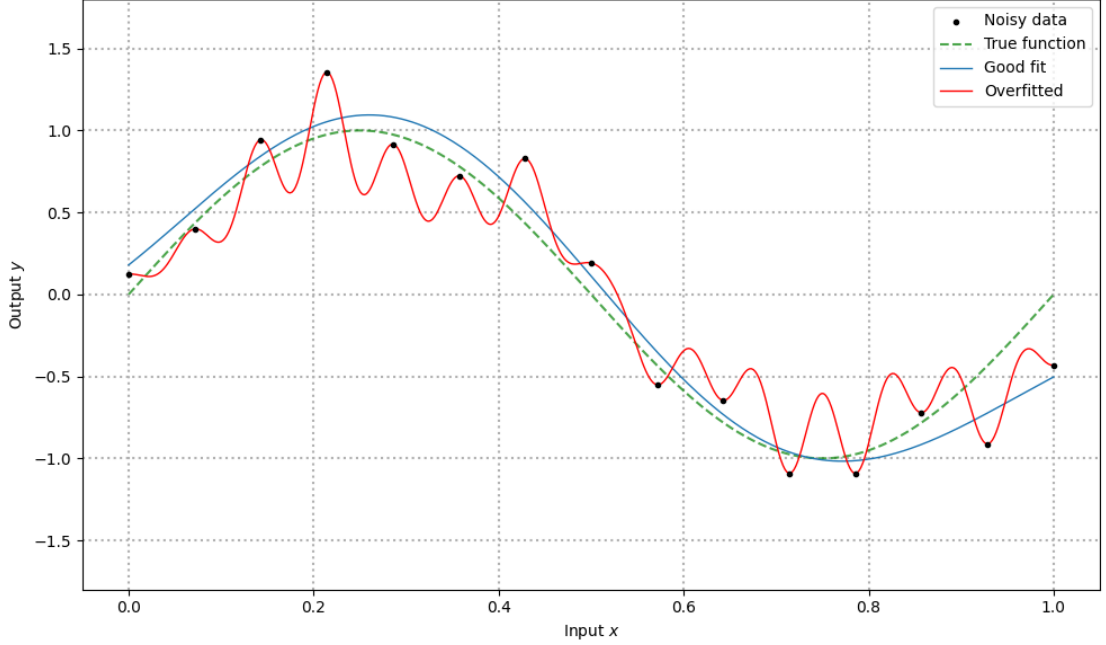


Figure 1.2: Example of overfitting. We do not impose any constraint on the regularity of the model, allowing it to fit perfectly the training data but with poor generalization capabilities, since its performance is highly noise-dependent.

where  $f$  belongs to a suitable function space.

The best predictor in  $\mathcal{H}$  is defined as the minimizer of the expected risk function. Theoretically the best possible predictor is the following:

**Definition 1.1.5.** We call Bayes Predictor the function:

$$f^*(x) = \operatorname{argmin}_{z \in \mathcal{Y}} \mathbb{E}_{y \sim \rho(y|x)}(l(y, z))$$

and the corresponding risk is called Bayes Risk, which is the best theoretically achievable risk<sup>1</sup>:

$$R(f^*) = \mathbb{E}_{x \sim \rho_X} \left( \inf_z \mathbb{E}_{y \sim \rho(y|x)}(l(z, y)) \right)$$

It is clear that we do not have access to  $\rho$ , otherwise we could (numerically) compute the function  $f^*$  and get a state-of-the-art predictor. Since we do not have  $\rho$ , we cannot even calculate  $R(\cdot)$ ; we have then to rely on the following:

<sup>1</sup>This can be easily seen, since  $d\rho_X$  is a positive measure.

**Definition 1.1.6.** Let  $(x_i, y_i)_{i=1, \dots, n} \in \mathcal{X} \times \mathcal{Y}$  be the known dataset, let  $f$  an estimator. We define the empirical risk as:

$$\hat{R}(f) = \frac{1}{n} \sum_{i=1}^n l(f(x_i), y_i)$$

The minimization of the empirical risk is a core idea behind the majority of machine learning tasks.

Following the introduction, it appears that we didn't specify properly the space of functions  $\mathcal{H}$ . That is a crucial decision since it affects both computational and analytical aspects of the learning problem. Indeed, if we chose  $\mathcal{H}$  to be "too complicated" we could obtain a small empirical risk, but it would be analytically and computationally not feasible and we could also encounter overfitting issues. On the contrary, if the space is "too simple" it would be easy to learn but it would probably encounter high training error, and it would fail to properly generalize. So it is important to choose a function space which is not too complex, but also not too simple, in a trade-off between computational feasibility and generalization ability. Here kernel methods come handy, they allow us to work in a high-dimensional space where the data is more likely to be on a linear subspace, without explicitly computing the coordinates of the data in this space. This is a very powerful tool, and it is at the heart of many machine learning algorithms.

The easiest way to visualize the power of kernel methods is through data classification, even though we will focus on regression.

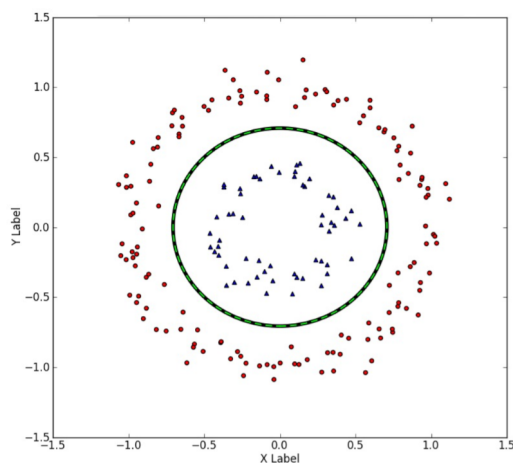


Figure 1.3: The separator is obviously circle-shaped but in general we do not want to impose any prior constraint on the function that the model can learn.

In Figure 1.3 we are trying to separate the two classes of data: to obtain a simple to learn linear model we lift our datasets into a higher dimensional space where they are likely to be linearly separable.

To do so we use the following lifting function:  $\phi(x) = (x_1 \ x_2 \ x_1^2 + x_2^2)^T$ . In such a way, in our new representation space, the data are linearly separable by a hyperplane of the form:  $z = c$ .

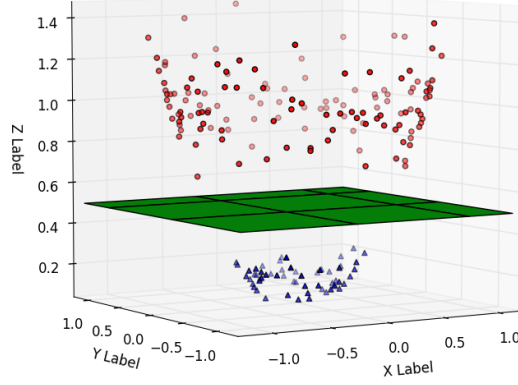


Figure 1.4: In this space the data are linearly separable even though the lifting function is non-linear.

And now, to get the separator in the original space we take the preimage of the function defining the hyperplane, and we get  $x_1^2 + x_2^2 = c$ , which defines a circle in  $\mathbb{R}^2$ . We show the final result in Figure 1.3. One could argue that we chose a convenient lifting function, and indeed we did that to get an explicit visualization; in reality there are general lifting functions, that we will soon call feature map, which would have provided the same circular boundary.

## 1.2 Kernel Methods

We start by defining a (positive-definite) kernel function:

**Definition 1.2.1.** Let  $\mathcal{X} \subseteq \mathbb{R}^d$  be the input space, we call kernel function, a symmetric function  $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$  s.t. for all  $n \in \mathbb{N}, x_1, \dots, x_n \in \mathcal{X}$  and  $c_1, \dots, c_n \in \mathbb{R}$ , we have:

$$\sum_{i=1}^n \sum_{j=1}^n c_i c_j k(x_i, x_j) \geq 0.$$

We could as well define the condition above as:

$$\langle c, Kc \rangle_{\mathbb{R}^n} \geq 0, \quad c \in \mathbb{R}^n, K_{ij} = k(x_i, x_j), K \in \mathbb{R}^{n \times n}$$

in such case we call  $K$  kernel matrix, and we ask for it to be positive semi-definite. Common examples are the linear kernel  $k(x, y) = x^T y$ , the min kernel  $k(x, y) = \min(x, y)$

over  $\mathcal{X} = \mathbb{R}_{>0}$  and in general every kernel that can be written as an inner product in some space, i.e.  $k(x, y) = \langle g(x), g(y) \rangle$  for some function  $g$ .

Now we have the following definition, which is the starting point to have a link between kernels and functions spaces:

**Definition 1.2.2.** Let  $\mathcal{X}$  be a set and  $\mathcal{H}$  be a function space from  $\mathcal{X}$  to  $\mathbb{R}$  forming a real Hilbert space with inner product  $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ . The function  $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$  is called a reproducing kernel (r.k.) of  $\mathcal{H}$  if

1.  $\mathcal{H}$  contains all functions of the form

$$\forall x \in \mathcal{X}, k_x : t \mapsto k(x, t)$$

2. For every  $x \in \mathcal{X}$  and  $f \in \mathcal{H}$  the reproducing property holds:

$$f(x) = \langle f, k_x \rangle_{\mathcal{H}}$$

If a r.k. exists, then  $\mathcal{H}$  is called a reproducing kernel Hilbert space (RKHS for short).

To further clarify the link we have the following theorem:

**Theorem 1.2.3.** A function  $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$  is a p.d. kernel if and only if it is a reproducing kernel of some Hilbert space  $\mathcal{H}$ .

**Theorem 1.2.4** (Aronszajn, 1950).  $k$  is a p.d. kernel on the set  $\mathcal{X}$  if and only if there exists a Hilbert space  $\mathcal{F}$  and a mapping  $\Phi : \mathcal{X} \rightarrow \mathcal{F}$ , such that, for any  $x, x'$  in  $\mathcal{X}$ :

$$k(x, x') = \langle \Phi(x), \Phi(x') \rangle_{\mathcal{F}}$$

Notice that, in general,  $\mathcal{F}$  and  $\mathcal{H}$  are not the same Hilbert space, but one can always chose  $\phi(x) \equiv k_x$  and it can be proven that the two spaces are isomorphic. Furthermore, as we are about to see, we do not need to know the explicit feature map, since most of the algorithms rely only on the "interaction between data", i.e. the value of the kernel function.

To see the link with learning theory we need to further expand the setting. First we assume our predictors to be linear with respect to the feature map, i.e. of the form:

$$\exists w \in \mathcal{H} \quad f_w : \mathcal{X} \rightarrow \mathbb{R}, \quad x \mapsto f_w(x) := \langle w, \phi(x) \rangle_{\mathcal{H}}$$

Note that the function is not linear on the data  $x$  itself, but on their representation in the feature space  $\mathcal{H}$ . Furthermore we consider the regularized empirical risk function to be of the form:

$$R(f_w, (x_i, y_i)_{i=1, \dots, n}) = \frac{1}{n} \sum_{i=1}^n L(f_w(x_i), y_i) + \lambda \|w\|^2$$

where  $L$  is the loss function (defined before) and  $\lambda \geq 0$  is the regularization parameter. Now we can state the following theorem, which allows us to restrict the optimal searching space.

**Theorem 1.2.5** (Representer theorem). Consider a feature map  $\phi : \mathcal{X} \rightarrow \mathcal{H}$ , let  $(x_1, \dots, x_n) \in \mathcal{X}^n$ , and consider a functional  $\Psi : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}$  which is strictly increasing with respect to the last variable. Then the infimum over  $w \in \mathcal{H}$  of

$$\Psi(\langle w, \phi(x_1) \rangle_{\mathcal{H}}, \dots, \langle w, \phi(x_n) \rangle_{\mathcal{H}}, \|w\|_{\mathcal{H}}^2)$$

can be obtained by restricting to a  $w$  in the span of  $\phi(x_1), \dots, \phi(x_n)$ ; that is, of the form

$$w = \sum_{i=1}^n \alpha_i \phi(x_i)$$

with  $\alpha \in \mathbb{R}^n$ .

This statement is the generalized representer theorem of Schölkopf, Herbrich and Smola (COLT 2001); the original result, in the spline setting, is due to Kimeldorf and Wahba (J. Math. Anal. Appl., 1971).

It comes straightforward that the loss function we defined above satisfies the hypothesis of the theorem, hence we can restrict our search space of optimal parameters to the space spanned by the feature map of the training data.

We conclude the paragraph presenting the notion of universality for a kernel:

**Definition 1.2.6.** A kernel  $k$  is said to be universal if for any choice of compact subset  $Z$  of the input space  $\mathcal{X}$ , the set  $K(Z) := \text{span}\{k_x : x \in Z\}$  is dense in  $C(Z)$ , the set of continuous functions over  $Z$ , w.r.t. the maximum norm.

So, if a kernel is universal, it can approximate arbitrarily well any continuous function over a compact set and so it can define a good model for the learning problem.

## 1.2.1 Translation Invariant Kernels

We now focus on a translation invariant kernels, which will be the starting point of this work.

**Definition 1.2.7.** A p.d. kernel  $k$  is called translation invariant if it satisfies the following property:

$$\forall x, y, t \in \mathcal{X} \quad k(x, y) = k(x+t, y+t) =: k^\Delta(x-y)$$

An example of translation invariant kernel is the Gaussian Kernel (or RBF) which is:

$$k(x, y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right), \quad \sigma > 0$$

or the Matérn Kernel:

$$k_\nu(x, y) = \frac{1}{2^{\nu-1}\Gamma(\nu)} \left(\frac{\sqrt{2\nu}\|x-y\|}{\rho}\right)^\nu K_\nu\left(\frac{\sqrt{2\nu}\|x-y\|}{\rho}\right)$$

where  $K_\nu$  denotes the modified Bessel function and  $\rho$  and  $\nu$  are positive parameters.

**Proposition 1.2.8.** The Gaussian kernel and the Matérn kernel are universal.

Translation invariant kernels are particularly convenient since the following theorem, due to Bochner, holds:

**Theorem 1.2.9** (Bochner, 1955). A continuous function  $k^\Delta$  is positive definite on  $\mathcal{X} \subseteq \mathbb{R}^d$  if and only if it is a Fourier transform of a positive finite measure  $\Lambda$  on  $\mathbb{R}^d$ , i.e.

$$k(x, y) = k^\Delta(x - y) = \int_{\mathbb{R}^d} e^{i\omega^T(x-y)} d\Lambda(\omega).$$

As we will see, due to computational limits, we have to rely on an approximation of the feature map, which usually is unknown. Bochner theorem allows us to tackle this issue. Indeed if we assume the kernel to be normalized, i.e.  $k(0) = 1$ , then we can rewrite the condition above as:

$$k^\Delta(u) = \mathbb{E}_{\omega \sim \Lambda}[e^{i\omega^T u}].$$

hence exploiting the law of large numbers, we can approximate the kernel function with Monte-Carlo approximation:

$$k^\Delta(x - y) \approx \frac{1}{M} \sum_{j=1}^M e^{i\omega_j^T(x-y)} = \left\langle \frac{1}{\sqrt{M}} e^{i\Omega^T x}, \frac{1}{\sqrt{M}} e^{i\Omega^T y} \right\rangle_{\mathbb{C}}, \quad \Omega = (\omega_1, \dots, \omega_M) \in \mathbb{R}^{d \times M}$$

where  $\omega_1, \dots, \omega_M$  are drawn according to  $\Lambda$ . Hence we can define:

**Definition 1.2.10.** Let  $k$  be a normalized translation invariant kernel. We define the random fourier feature map as:

$$\phi_{RFF}(x) := \frac{1}{\sqrt{M}} e^{i\Omega^T x} = \frac{1}{\sqrt{M}} \begin{pmatrix} e^{i\omega_1^T x} \\ \vdots \\ e^{i\omega_M^T x} \end{pmatrix}, \quad x \in \mathcal{X},$$

with  $\Omega = (\omega_1, \dots, \omega_M) \in \mathbb{R}^{d \times M}$  where each  $\omega_j$  is drawn from  $\Lambda$ , the probability distribution on  $\mathbb{R}^d$  associated with the kernel  $k$ . The corresponding kernel function is then:

$$k_{RFF}(x, y) = \langle \phi_{RFF}(x), \phi_{RFF}(y) \rangle_{\mathbb{C}} = \frac{1}{M} \sum_{i=1}^M e^{i\omega_i^T(x-y)}$$

which is shift invariant and an unbiased estimator for the original kernel function.

Since we will deal only with real kernels, we consider only the real part of the Fourier function:

$$\Re(e^{i\Omega^T x}) = \cos(\Omega^T x),$$

and so the real part of the approximation kernel is:

$$\Re(k_{RFF}(x, y)) = \frac{1}{M} \sum \cos(\omega_i(x - y))$$

to have an approximation of it we consider the following feature map:

$$\phi_{real}(x) := \frac{1}{\sqrt{2M}} \begin{pmatrix} \cos(\omega_1^T x) \\ \vdots \\ \cos(\omega_M^T x) \\ \sin(\omega_1^T x) \\ \vdots \\ \sin(\omega_M^T x) \end{pmatrix} \in \mathbb{R}^{2M}$$

in such a way we have:

$$\begin{aligned} \langle \phi_{real}(x), \phi_{real}(y) \rangle &= \frac{1}{2M} \sum_{i=1}^{2M} \cos(\omega_i^T x) \cos(\omega_i^T y) + \sin(\omega_i^T x) \sin(\omega_i^T y) \\ &= \frac{1}{2M} \sum \cos(\omega_i^T (x - y)) \end{aligned}$$

this is an unbiased estimator of (the real part of the) initial kernel. Usually the feature map above is the common choice for RFFs applications. But, as we previously mentioned, we will deal with memory constrained applications, hence it is important for our data representation to occupy as little space as possible, hence we work with the following modification:

**Definition 1.2.11.** Let  $x \in \mathcal{X}$ ,  $M \in \mathbb{N}_{>0}$  and  $\omega_1, \dots, \omega_M$  sampled from  $\Lambda$ , then we define:

$$\phi_M(x) := \sqrt{\frac{2}{M}} \cos(\Omega^T x + \xi) = \sqrt{\frac{2}{M}} \begin{pmatrix} \cos(\omega_1^T x + \xi_1) \\ \vdots \\ \cos(\omega_M^T x + \xi_M) \end{pmatrix}$$

where  $\xi = (\xi_1, \dots, \xi_M)^T$  and  $\xi_i \stackrel{i.i.d.}{\sim} \text{Unif}[0, 2\pi]$ ,  $\xi_j \perp \omega_h \forall h, j \in [M]$ .

In such a way, exploiting the independence of random variables, we have:

$$\begin{aligned} \mathbb{E}(\phi_M(x)_i \phi_M(y)_i) &= \frac{2}{M} \mathbb{E}_{\substack{\omega \sim \Lambda \\ \xi \sim \mathcal{U}}} [\cos(\omega_i^T x + \xi_i) \cos(\omega_i^T y + \xi_i)] \\ &= \frac{2}{M} \mathbb{E} \left[ \frac{1}{2} \cos(\omega^T (x - y)) + \frac{1}{2} \cos(\omega^T (x + y) + 2\xi) \right] \\ &= \frac{1}{M} \mathbb{E} [\cos(\omega^T (x - y))] + \mathbb{E}_\omega \mathbb{E}_\xi [\cos(\omega^T (x + y) + 2\xi) \mid \omega] \\ &= \frac{1}{M} \mathbb{E}(\cos(\omega^T (x - y))) \\ &= \frac{1}{M} k(x, y) \end{aligned}$$

and so:

$$\mathbb{E}(\langle \phi_M(x), \phi_M(y) \rangle) = k(x, y) = k^\Delta(x - y)$$

hence we have an unbiased estimator of  $k$  which occupies half the space of the previous one.

### 1.2.2 Feature map Quantization

As previously stated, our focus will be on reducing the amount of memory needed to store the feature map used in learning algorithms. To do so, instead of storing the full precision feature map  $z(x) := \phi_M(x) \in \mathbb{R}^M$ , which usually requires 32-bit or 64-bit floating-point precision, we can store a quantized version of it,  $\tilde{z}(x) \in \mathbb{R}^M$ , which can be stored using a reduced number of bits  $b$ . We use the following procedure:

1. Given  $M$  and  $b$ , we divide the interval  $\left[-\sqrt{\frac{2}{M}}, \sqrt{\frac{2}{M}}\right]$  into  $2^b - 1$  equal sub-intervals, defined by  $2^b$  quantization levels  $q_0 < q_1 < \dots < q_{2^b-1}$ .
2. For each coordinate  $j \in \{1, \dots, M\}$  and each  $x \in \mathbb{R}^d$ , assuming  $z_j(x) \in [q_k, q_{k+1}]$ , we define the stochastic quantization operator  $Q(\cdot)$  as:

$$Q(z_j(x)) = \begin{cases} q_k & \text{with probability } \frac{q_{k+1} - z_j(x)}{q_{k+1} - q_k} \\ q_{k+1} & \text{with probability } \frac{z_j(x) - q_k}{q_{k+1} - q_k} \end{cases} \quad (1.4)$$

By defining  $(\phi_M^q(x))_j := Q(z_j(x))$ , this formulation ensures that the quantizer is unbiased, meaning  $\mathbb{E}[Q(z_j(x)) \mid z_j(x)] = z_j(x)$ . Consequently, the quantized feature map preserves the expected inner product properties of the full-precision map while drastically reducing the memory footprint.

In particular we choose  $b = 1$ , i.e. we work with the lowest possible amount of bits, meaning that we quantize each feature on the rescaled unit hypercube  $\left[-\sqrt{\frac{2}{M}}, \sqrt{\frac{2}{M}}\right]^M$ . Another common procedure to perform quantization is by taking the sign of each feature, i.e. we quantize each feature, i.e.

$$Q_s(z_j(x)) = \sqrt{\frac{2}{M}} \text{sign}(z_j(x))$$

this approach is computationally more feasible, since we do not need to perform the sampling of the quantization levels and neither the exact feature, since the cosine wave is periodic. However, analytically it is more challenging. Indeed, in expectation over the draw of  $\Omega$  and  $\xi$  we have [16]:

$$\forall x, y \quad \mathbb{E}(\langle Q_s(z(x)), Q_s(z(y)) \rangle) = \sum_{\substack{h \in \mathbb{Z} \\ h \text{ odd}}} |S_h|^2 k^\Delta(h(x - y)), \quad S_h := \frac{2}{\pi h} (-1)^{(h-1)/2} \quad (1.5)$$

which is not the original kernel  $k(x, y)$ . However, since it is a series of kernels, it is still a kernel and its RKHS includes the one spawned by the original kernel map [6]. For this reason its approximation capabilities are better than our procedure and we leave its study for future works.

### 1.3 A note on the analysis of Chapter 3

The analysis of Chapter 3 assumes that the noise variables  $\gamma = (\gamma_0, \dots, \gamma_{T-1})$  are re-drawn independently at every step. This assumption is relevant in our analysis, since it gives  $\mathbb{E}[H_t | \mathcal{F}_t] = 0$  in (3.49), hence both the factor  $1/b$  in the variance and the decomposition of  $\delta_T$ .

What this regime saves is arithmetic, not storage. The rounding probabilities are determined by  $\cos(\omega_j^\top x_i + \xi_j)$ , so a new noise variable requires the full-precision value of the coordinate. To avoid this issue we can think of the noise as fixed for each sample for all the training time, i.e.  $\gamma_{t,i} = \gamma_{0,i}$  for all  $t$ . In this case the features are computed once at the beginning of the training, then they are kept in memory. Hence the representation is stored only in the *frozen* scheme, in which the features are quantized once:

$$S_0^q := \begin{pmatrix} \phi_M^q(x_1, \gamma_{0,1})^\top \\ \vdots \\ \phi_M^q(x_n, \gamma_{0,n})^\top \end{pmatrix} = \sqrt{2/M} \Sigma_0, \quad \Sigma_0 \in \{\pm 1\}^{n \times M}, \quad (1.6)$$

and kept on  $nM$  bits compared to the  $32nM$  of full precision. Training samples from the matrix  $S_0^q$  and so we have:

$$w_0^{qf} := 0, \quad w_{t+1}^{qf} = w_t^{qf} - \frac{\mu}{b} \sum_{i \in B_t} (\langle (S_0^q)_i, w_t^{qf} \rangle - y_i) (S_0^q)_i^\top, \quad (1.7)$$

and  $\Omega, \xi, X$  are no longer needed. We compare (1.7) with (3.25) in two regimes, neither covered by the analysis, for opposite reasons.

- **Multi-epoch**,  $Tb \gg n$ , mini-batches drawn with replacement as in [4]. In this regime, the independence of the batches is preserved, but the noise variable is reused multiple times. Hence  $w_t^{qf}$  is correlated with the  $\gamma_{0,i}$  entering step  $t$  and (3.49) fails. In this regime we evaluate the effect of independence of the noise term. Comparing our estimator with its frozen counterpart we can establish if fixing the noise term is impacting the performance in a significant way.
- **Single-pass**,  $Tb \leq n$ , samples visited in a random permutation. Each noise variable is used once, so (3.25) and (1.7) coincide in law and (3.49) holds. But sampling is now without replacement, so the mini-batches are neither independent nor identically distributed: the identification of the law of  $P'_k$  with that of  $\Pi'_s$  in Propositions 3.4.2 and 3.4.6, and the bound of [4] used for (3.38), lose their justification.

On the contrary, in this regime we evaluate the effect of the lack of independence of the mini-batches. Comparing our estimator with its frozen counterpart we can establish if sampling without replacement, hence maintaining independence of the noise term but not of the batches, impacts performance in a significant way.

The frozen estimator is therefore covered in no parameter configuration, and closing this gap is the a question left open here. Empirically the two estimators agree in both regimes, see Figures 4.9–4.8. This could suggest a relaxation of the hypothesis in the theoretical analysis is possible. And, in a such way, the possibility to use a modification of our estimator in a low-memory setting.



## Related Work

Our contribution sits at the intersection of three lines of research: random feature approximations of kernels and their statistical analysis (Sections 2.1 and 2.2), learning under a memory budget (Section 2.3), and quantization of feature maps (Sections 2.4 and 2.5). We review them in turn, and position our contribution with respect to them in Section 2.6.

### 2.1 Random Features for Kernel Approximation

Kernel methods scale poorly with the sample size  $n$ : forming and factorizing the kernel matrix costs  $O(n^2)$  memory and  $O(n^3)$  time. Two families of randomized approximations dominate. *Nyström* methods (Williams and Seeger, NIPS 2000) project the full kernel matrix onto the span of a subsample of its columns, so that only the sampled columns and a small factor need to be stored. In [14] it is proved that, under standard assumptions, Nyström kernel ridge regression with uniform subsampling achieves optimal statistical learning rates, and that the number of sampled columns itself acts as a regularization parameter.

*Random Fourier features* (RFFs) [12] do not subsample the data at all: they use Bochner’s theorem [6] to write a normalized translation invariant kernel as  $k^\Lambda(x - y) = \mathbb{E}_{\omega \sim \Lambda}[e^{i\omega^\top(x-y)}]$  and approximate the expectation by Monte Carlo, with a map  $\phi_M : \mathcal{X} \rightarrow \mathbb{R}^M$  with  $\langle \phi_M(x), \phi_M(y) \rangle \approx k(x, y)$ . The authors provide uniform approximation guarantees for common kernel functions over compact sets, and [13] analyze the resulting function class as a random one-hidden-layer network.

As we will see in Chapter 1, two real-valued variants of the RFFs map coexist in the literature: the *stacked* map  $\tilde{z}(x) = M^{-1/2}(\cos(\Omega^\top x), \sin(\Omega^\top x))$  and the *dithered* map  $\check{z}(x) = \sqrt{2/M}\cos(\Omega^\top x + \xi)$ ,  $\xi \sim \text{Unif}[0, 2\pi]^M$ . [17] compute the variance of both and show that for the Gaussian kernel  $\tilde{z}$  is uniformly lower-variance. The map we quantize is the dithered one. The trade-off is not simply in its favour: at equal number of frequencies the dithered map stores  $M$  coordinates instead of  $2M$ , but in full precision one needs roughly twice as many dithered features to match the accuracy of the stacked

map, so the two are comparable at equal memory. What singles out the dithered map here is quantization itself: each coordinate is a single cosine taking values in the fixed interval  $[-\sqrt{2/M}, \sqrt{2/M}]$ , so the one-bit stochastic rounding of Section 2.4 is defined coordinatewise on a known range and its noise is independent across coordinates. Both facts are used in Chapter 3, where they yield a quantization perturbation that is diagonal in expectation.

A parallel line of work reduces the cost of *generating* the features rather than storing them: Fastfood [9] replaces the dense Gaussian matrix  $\Omega$  by a product of Hadamard and diagonal matrices, giving  $O(M \log d)$  time and  $O(M)$  memory; circulant and other structured projections [19] reduce the storage of the random projection matrix  $\Omega \in \mathbb{R}^{d \times M}$  to  $O(M)$ . These techniques are independent of feature quantization and can be combined with it; indeed we can retain the approximation of the projection matrix and then approximate the outcome by quantization procedures.

## 2.2 Statistical Analysis of Learning with Random Features

Early analyses decoupled kernel approximation from learning, bounding  $\|k - k_M\|_\infty$  and paying an  $O(1/\sqrt{M})$  price in the risk, which suggests  $M \approx n$  features. Sharper analyses study the excess risk directly. In [15] is proved that, for kernel ridge regression  $M \approx \sqrt{n} \log n$  plain random features already suffice to achieve an excess risk of order  $O(n^{-1/2})$ . This rate is *minimax optimal* in the sense of [3]: over the class of distributions considered, no estimator can achieve a smaller worst-case excess risk, so the estimator is as accurate as is information-theoretically possible. Furthermore, [2] obtained matching upper and lower bounds showing that plain RFFs need  $M = \mathcal{O}(n/\lambda)$  features for the approximate kernel matrix to be a  $\Delta$ -spectral approximation of  $K + \lambda I$ , that is, for

$$(1 - \Delta)(K + \lambda I) \preceq \tilde{K} + \lambda I \preceq (1 + \Delta)(K + \lambda I)$$

to hold in the positive semi-definite order. This is a relative guarantee on the whole spectrum, and it is what allows the ridge regression solution computed with  $\tilde{K}$  to be compared to the exact one. The core work behind ours is [4], where an optimal convergence rate is proven for mini-batch SGD with RFFs.

The characterization of which kernels induce a rich enough hypothesis space is due to [11]. We call *universal* the kernels capable of approximating any continuous map over a (compact) set, i.e.  $\overline{\text{span}}\{k_x : x \in Z\} = C(Z)$  for every compact  $Z$ . For translation invariant kernels on  $\mathbb{R}^d$ , universality depends only on the support of the spectral measure  $\Lambda$  appearing in Bochner's theorem (Chapter 1): two such kernels whose spectral measures share the same support are either both universal or both not (their Theorem 12). A simple sufficient condition is that  $\text{supp } \Lambda$  have positive Lebesgue measure (their Proposition 15). This is what guarantees that the Gaussian and Matérn kernels used in our experiments are universal, since their spectral densities are strictly positive on all of  $\mathbb{R}^d$ .

## 2.3 Low-Memory Kernel Approximation

The results of Section 2.2 are stated in terms of the number of features  $M$ , which is the natural currency when time is the binding constraint. Once memory is the binding constraint the comparison has to be redone in bytes rather than in features, and the ranking of the methods changes.

Considering now a *memory* rather than a time budget, [18] showed that at equal number of features Nyström generalizes better than RFFs when the spectrum decays quickly, but [20] demonstrated that the comparison reverses at equal memory, since Nyström must additionally store an  $M \times d$  projection matrix plus the  $M \times M$  factor, whereas structured or quantized RFFs do not.

## 2.4 Quantization of Random Features

Two complementary paradigms exist for quantizing random periodic features.

**Deterministic (one-bit / universal) quantization.** Replacing the cosine in the RFFs map by a binary periodic function  $f$  yields features taking values in  $\{-1, +1\}^M$ , which can therefore be stored on one bit per coordinate. The price to pay is a bias at the kernel level: by Proposition 2.5 of [16], the symmetric random periodic feature (RPF) inner product converges to a *scale mixture*  $\kappa_{f,f}^\Delta(u) = \sum_{k \in \mathbb{Z}} |F_k|^2 \kappa^\Delta(ku)$  of the target kernel, where  $F_k$  are the Fourier coefficients of the periodic map. Taking  $f = q$ , with  $q(t) = \text{sgn}(\cos t)$  the square wave used as a one-bit quantizer, this distorts the Gaussian kernel visibly near the origin, which is precisely the behaviour we reproduce in Figures 4.1 and 4.2. Despite this, the mixture kernel  $\kappa_{q,q}^\Delta$  is itself a kernel, and subtracting from it a small enough positive multiple of  $\kappa^\Delta$  leaves a sum of positive definite kernels; its RKHS therefore *contains* the RKHS of the target kernel  $\kappa^\Delta$ , and the extra capacity can yield better generalization performance.

The work [16] proposes to break the symmetry: with  $\varphi(x) = f(\Omega^\top x + \xi)$  and  $\psi(y) = g(\Omega^\top y + \xi)$ , the expected kernel is  $\kappa_{f,g}^\Delta(u) = \sum_k F_k \overline{G_k} \kappa^\Delta(ku)$  (their Proposition 3.1), and the choice  $f = q$ ,  $g = \cos$  recovers the *undistorted* kernel up to the factor  $2/\pi$ . Since  $q$  is discontinuous, uniform bounds over infinite compact sets cannot rely on Lipschitz continuity; the authors introduce the *mean Lipschitz smoothness* property, which is satisfied by square waves, and obtain uniform error bounds — correcting in passing an error in the earlier  $T$ -part Lipschitz analysis. The scheme is intrinsically asymmetric: the database is stored in quantized form, while the query against which it is compared is kept in full precision. In the memory-constrained setting we consider we therefore do not use it, since one of the two maps must be represented in full precision. This does not rule out an asymmetric variant of our own procedure — learning on quantized features and then performing some form of asymmetric reconstruction at inference time — which we do not pursue here.

**Stochastic (unbiased) quantization.** The alternative is to round each feature at random so that the quantizer is unbiased. [20] introduce low-precision RFFs (LP-RFFs): each coordinate of  $z(x) = \sqrt{2/M} \cos(\Omega^\top x + \xi) \in [-\sqrt{2/M}, \sqrt{2/M}]$  is rounded to one of  $2^b$  levels with probabilities making  $\mathbb{E}[Q(z_j) | z_j] = z_j$ . By their Proposition 7 the noise variance obeys  $\mathbb{E}[C_{ij}^2 | Z_{ij}] \leq \delta_b^2/M$  with  $\delta_b^2 := 2/(2^b - 1)^2$ , where  $Z \in \mathbb{R}^{n \times M}$  collects the full-precision features of the dataset and  $C$  the corresponding rounding errors, so that  $Z + C$  is the quantized feature matrix. Consequently  $\mathbb{E}[(Z + C)(Z + C)^\top] = K + s_b I_n$  with  $0 \leq s_b \leq \delta_b^2$ : quantization acts, in expectation, as an additional ridge on the kernel matrix. Their main guarantee (Theorem 2) is that  $\tilde{K} + \lambda I$ , where  $\tilde{K}$  is the approximated kernel matrix, remains a  $(\Delta_1, \Delta_2)$ -spectral approximation of  $K + \lambda I$  as long as  $\Delta_2 \geq \delta_b^2/\lambda$ , i.e. quantization is harmless whenever the quantization noise is small relative to the regularization. Empirically LP-RFFs match full-precision RFFs and Nyström with 3–10 $\times$  and 50–460 $\times$  less memory. Their motivation is memory during *training*; inference is performed in full precision.

Our setting inherits the stochastic rounding of [20] at the extreme precision  $b = 1$ , but differs from it in what is analyzed. Zhang et al. bound a kernel-matrix approximation quantity. We instead propagate the quantization noise through the SGD recursion in the population  $L^2(\rho_{\mathcal{X}})$  norm, obtaining an explicit bias/variance decomposition of  $\|f_T^q - f_{\mathcal{H}}\|_\rho^2$  that composes with the bound of [4].

## 2.5 Low Precision in Machine Learning

Outside kernel methods, low-precision arithmetic is a standard tool for reducing memory and compute. Stochastic rounding was popularized by [8], who trained networks in 16-bit fixed point; binary and ternary weight networks [5] push this further by quantizing the parameters themselves. A parallel line quantizes the *gradients* rather than the weights, as in QSGD [1], where unbiasedness of the quantizer is exactly what preserves the convergence rate of SGD—the same mechanism we exploit in Chapter 3.

## 2.6 Positioning

Summarizing, [4] analyze SGD with full-precision random features; [20] quantize the features but analyze a kernel-matrix surrogate; [16] analyze one-bit periodic features, but only in terms of how well the quantized inner product approximates the target kernel, and without following the features through a training procedure. The present work studies 1-bit stochastically quantized random features *inside* a mini-batch SGD least-squares procedure and controls the resulting excess risk.

## Main results

This chapter contains the theoretical contribution of the thesis. After fixing the notation and the operators associated with the quantized feature map  $\phi_M^q$ , we show that stochastic rounding perturbs the covariance operator by a diagonal term of order  $1/M$ , and we propagate this perturbation through the mini-batch SGD recursion, bounding separately the resulting bias and variance. No explicit penalty is added anywhere: regularization is implicit, and is controlled by the step-size, the number of iterations and the number of features.

### 3.1 Notation and setting

We start by setting the notation we will be using in this work. Let  $x \in \mathcal{X} \subset \mathbb{R}^d$ . As stochastic feature map we consider:

$$\phi_M^q(x, \gamma) = \sqrt{\frac{2}{M}} (\cos(\Omega^T x + \xi) + \bar{\eta}(x, \gamma)) = \phi_M(x) + \eta(x, \gamma) \quad (3.1)$$

where  $\Omega \sim \Lambda^M$ ,  $\xi \sim \text{Unif}[0, 2\pi]^M$ .

The noise term  $\gamma$  belongs to a probability space  $(\Gamma, \mathcal{G}, \mathbb{Q})$  and we have mutual independence between  $x, \Omega, \xi, \gamma$ . From a practical standpoint we can consider  $\gamma$  as an  $m$ -dimensional vector, living in the centered  $m$ -dimensional cube with size equal to 2 endowed with uniform probability.

Furthermore  $\bar{\eta}$  is a random vector in  $\mathbb{R}^M$  whose entries  $\bar{\eta}_j$ , for fixed values of  $\omega_j, \xi_j, x$ , are independent and satisfies:

$$\begin{aligned} \mathbb{Q}(\bar{\eta}_j + \cos(\omega_j^T x + \xi_j) = 1) &= \frac{1 + \cos(\omega_j^T x + \xi_j)}{2} \\ \mathbb{Q}(\bar{\eta}_j + \cos(\omega_j^T x + \xi_j) = -1) &= \frac{1 - \cos(\omega_j^T x + \xi_j)}{2} \end{aligned} \quad (3.2)$$

from which we can note that for fixed values of  $x, \Omega, \xi$ :

$$\mathbb{E}_\gamma [\bar{\eta}_j + \cos(\omega_j^T x + \xi_j)] = \cos(\omega_j^T x + \xi_j) \quad (3.3)$$

This design minimizes the memory footprint required to store the approximate RFF representation, as storing each dimension of the quantized map requires only 1 bit.

We define the following standard deterministic operators:

$$\begin{aligned} S_M : \mathbb{R}^M &\rightarrow L^2(\mathcal{X}) \\ S_M(w)(\cdot) &= \langle w, \phi_M(\cdot) \rangle_{\mathbb{R}^M} \end{aligned} \quad (3.4)$$

and its adjoint operator  $S_M^*$ :

$$\begin{aligned} S_M^* : L^2(\mathcal{X}) &\rightarrow \mathbb{R}^M \\ S_M^*(f)_i &= \langle f, (\phi_M)_i \rangle_{L^2(\mathcal{X})} = \int_{\mathcal{X}} f(x) \phi_M(x)_i d\rho_{\mathcal{X}}(x) \quad i = 1, \dots, M. \end{aligned} \quad (3.5)$$

From these, we construct the integral operator  $L_M := S_M \circ S_M^*$  associated with the exact kernel  $k_M$ :

$$\begin{aligned} L_M : L^2(\mathcal{X}) &\rightarrow L^2(\mathcal{X}) \\ L_M(f)(x) &= \langle \langle f, \phi_M \rangle_{L^2(\mathcal{X})}, \phi_M(x) \rangle_{\mathbb{R}^M} = \int_{\mathcal{X}} f(x') \langle \phi_M(x'), \phi_M(x) \rangle_{\mathbb{R}^M} d\rho_{\mathcal{X}}(x') \\ &= \int_{\mathcal{X}} k_M(x, x') f(x') d\rho_{\mathcal{X}}(x') \end{aligned} \quad (3.6)$$

and the corresponding finite-dimensional covariance operator  $C_M := S_M^* \circ S_M$ :

$$\begin{aligned} C_M : \mathbb{R}^M &\rightarrow \mathbb{R}^M \\ C_M(w) &= \langle \langle w, \phi_M(\cdot) \rangle_{\mathbb{R}^M}, \phi_M \rangle_{L^2(\mathcal{X})} = \int_{\mathcal{X}} \langle w, \phi_M(x) \rangle_{\mathbb{R}^M} \phi_M(x) d\rho_{\mathcal{X}}(x) \end{aligned} \quad (3.7)$$

In the learning procedure we will also consider the empirical versions of these operators, defined through  $\hat{S}_M : \mathbb{R}^M \rightarrow \mathbb{R}^n$  as follows over a dataset  $(x_i)_{i=1}^n$ :

$$\hat{S}_M^T v = \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi_M(x_i) v_i, \quad \hat{L}_M := \hat{S}_M \hat{S}_M^T, \quad \hat{C}_M := \hat{S}_M^T \hat{S}_M \quad (3.8)$$

We will denote the stochastic operators, i.e., the ones using  $\phi_M^q$ , by  $S_M^q, (S_M^q)^*, L_M^q, C_M^q$  and their empirical counterparts  $\hat{S}_{M,q}, \hat{S}_{M,q}^T, \hat{L}_{M,q}, \hat{C}_{M,q}$ . To clarify the definitions of

$L_M^q$  and  $C_M^q$ , we first explicitly provide the expression of the adjoint operator  $(S_M^q)^*$ :

$$\begin{aligned} S_M^q : \mathbb{R}^M &\rightarrow L^2(\mathcal{X} \times \Gamma), \\ S_M^q(w) &= \langle w, \phi_M^q(\cdot, \cdot) \rangle_{\mathbb{R}^M} \\ (S_M^q)^* : L^2(\mathcal{X} \times \Gamma) &\rightarrow \mathbb{R}^M, \end{aligned} \quad (3.9)$$

$$(S_M^q)^*(f) = \int_{\mathcal{X} \times \Gamma} f(x, \gamma) \phi_M^q(x, \gamma) d\rho_{\mathcal{X}}(x) d\mathbb{Q}(\gamma) \quad (3.10)$$

$$\begin{aligned} L_M^q : L^2(\mathcal{X} \times \Gamma) &\rightarrow L^2(\mathcal{X} \times \Gamma), \\ L_M^q(f) &= \langle \langle f, \phi_M^q \rangle_{L^2(\mathcal{X} \times \Gamma)}, \phi_M^q(\cdot, \cdot) \rangle_{\mathbb{R}^M} \end{aligned} \quad (3.11)$$

$$\begin{aligned} C_M^q : \mathbb{R}^M &\rightarrow \mathbb{R}^M, \\ C_M^q &:= (S_M^q)^* S_M^q \end{aligned} \quad (3.12)$$

Notice that these continuous population operators no longer work in  $L^2(\mathcal{X})$ , but in  $L^2(\mathcal{X} \times \Gamma, \rho_{\mathcal{X}} \otimes \mathbb{Q})$ , which is a space of stochastic functions. We will write  $L^2(\mathcal{X} \times \Gamma)$  for short.

For a given dataset  $(x_i, y_i)_{i=1}^n$  and noise realizations  $\gamma_i$ , we define the empirical versions. Strictly speaking, these are random matrices defined on the joint probability space of the data and the quantization noise. However, to keep notations simple where no ambiguity arises, we treat them as standard linear operators between finite-dimensional vector spaces:

$$\hat{S}_M^q : \mathbb{R}^M \rightarrow \mathbb{R}^n, \quad \hat{S}_M^q(w) = \frac{1}{\sqrt{n}} (\langle w, \phi_M^q(x_i, \gamma_i) \rangle_{\mathbb{R}^M})_{i=1}^n \quad (3.13)$$

$$(\hat{S}_M^q)^T : \mathbb{R}^n \rightarrow \mathbb{R}^M, \quad (\hat{S}_M^q)^T(v) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi_M^q(x_i, \gamma_i) v_i \quad (3.14)$$

$$\hat{L}_M^q : \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad \hat{L}_M^q = \hat{S}_M^q (\hat{S}_M^q)^T \quad (3.15)$$

$$\hat{C}_M^q : \mathbb{R}^M \rightarrow \mathbb{R}^M, \quad \hat{C}_M^q = (\hat{S}_M^q)^T \hat{S}_M^q \quad (3.16)$$

Since our aim is to compare deterministic functions, we will restrict  $L^2(\mathcal{X} \times \Gamma)$  to the image of the embedding  $i : L^2(\mathcal{X}) \hookrightarrow L^2(\mathcal{X} \times \Gamma)$  and work with the expected values of the above operators; under this assumption we have:

$$(S_M^q)^* i(f) = \int_{\mathcal{X}} f(x) \int_{\Gamma} \phi_M^q(x, \gamma) d\mathbb{Q}(\gamma) d\rho_{\mathcal{X}}(x) = \int_{\mathcal{X}} f(x) \phi_M(x) d\rho_{\mathcal{X}}(x) = S_M^* f$$

we can further exploit the independence of  $w$  on  $\gamma$  to get:

$$\mathbb{E}_{\gamma} [S_M^q(w)(x, \gamma)] = \langle w, \mathbb{E}_{\gamma} [\phi_M^q(x, \gamma)] \rangle_{\mathbb{R}^M} = \langle w, \phi_M(x) \rangle_{\mathbb{R}^M} = S_M(w)(x)$$

And we have:

$$\begin{aligned}
C_M^q(w) &= (S_M^q)^*(S_M^q(w)) \\
&= \int_{\mathcal{X} \times \Gamma} (S_M^q(w)(x, \gamma)) \phi_M^q(x, \gamma) d\rho_{\mathcal{X}}(x) d\mathbb{Q}(\gamma) \\
&= \int_{\mathcal{X} \times \Gamma} \langle \phi_M^q(x, \gamma), w \rangle_{\mathbb{R}^M} \phi_M^q(x, \gamma) d\rho_{\mathcal{X}}(x) d\mathbb{Q}(\gamma) \\
&= \int_{\mathcal{X}} \left( \int_{\Gamma} \phi_M^q(x, \gamma) \phi_M^q(x, \gamma)^T d\mathbb{Q}(\gamma) \right) w d\rho_{\mathcal{X}}(x) \\
&= \left( \int_{\mathcal{X}} \phi_M(x) \phi_M(x)^T d\rho_{\mathcal{X}}(x) \right) w + \left( \int_{\mathcal{X}} D(x) d\rho_{\mathcal{X}}(x) \right) w \\
&= C_M(w) + \bar{D}w = (C_M + \bar{D})w
\end{aligned} \tag{3.17}$$

where we used that the quantization noise has zero mean given  $x$ , hence the mixed terms disappear, and  $\bar{D} := \mathbb{E}_x(D(x))$ . Where:

$$D(x) := \mathbb{E}_{\gamma} [\eta(x, \gamma) \eta(x, \gamma)^T] \tag{3.18}$$

we notice that  $D(x)$  is a diagonal matrix, since we have independence of each quantization noise given  $x$ , thus:

$$D(x)_{i,i} = \mathbb{E}_{\gamma} [\eta_i(x, \gamma)^2] \leq \frac{2}{M} \tag{3.19}$$

where the last inequality comes from Proposition 7 in [20]. Furthermore, assuming the continuity of the distribution of the data, we have:

$$D(x)_{i,i} > 0 \quad a.s. \tag{3.20}$$

Moving now to the empirical counterpart, let  $\eta_i(\cdot) := \eta(x_i, \cdot)$  denote the quantization noise vector realized at the  $i$ -th sample. We have:

$$\mathbb{E}_{\gamma}(\phi_M^q(x_i, \gamma_i)) = \phi_M(x_i) \Rightarrow \mathbb{E}_{\gamma}(\hat{S}_M^q) = \hat{S}_M \tag{3.21}$$

For the empirical covariance matrix, the noise expands quadratically:

$$\begin{aligned}
\mathbb{E}_{\gamma}[\hat{C}_M^q] &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\gamma_i} [\phi_M^q(x_i, \gamma_i) \phi_M^q(x_i, \gamma_i)^T] \\
&= \frac{1}{n} \sum_{i=1}^n (\phi_M(x_i) \phi_M(x_i)^T + D(x_i)) = \hat{C}_M + \hat{D}_n, \quad \hat{D}_n = \frac{1}{n} \sum_{i=1}^n D(x_i)
\end{aligned} \tag{3.22}$$

where  $\hat{D}_n = \frac{1}{n} \sum_{i=1}^n D(x_i)$ . Analogously, for the kernel matrix elements  $(\hat{L}_M^q)_{i,j}$ :

$$\begin{aligned}
\mathbb{E}_{\gamma}[(\hat{L}_M^q)_{i,j}] &= \frac{1}{n} \phi_M(x_i)^T \phi_M(x_j) + \frac{\delta_i^j}{n} \mathbb{E}_{\gamma_i} [\eta_i^T \eta_i] \\
&= (\hat{L}_M)_{i,j} + \frac{\delta_i^j}{n} \mathbb{E}_{\eta_i} [\text{Tr}(\eta_i \eta_i^T)] \\
&= (\hat{L}_M)_{i,j} + \frac{\delta_i^j}{n} \text{Tr}(D(x_i))
\end{aligned}$$

hence:

$$\mathbb{E}_\eta [\hat{L}_M^q] = \hat{L}_M + \hat{\Delta}_L \quad (3.23)$$

where  $\hat{\Delta}_L := \text{diag} \left( \frac{\text{Tr}(D(x_1))}{n}, \dots, \frac{\text{Tr}(D(x_n))}{n} \right)$ .

## 3.2 The algorithm

The aim now is to investigate the learning speed of the function recovered through SGD with respect to the theoretical optima in the RKHS:  $f_{\mathcal{H}} := \arg \min_{f \in \mathcal{H}} \mathcal{E}(f)$  where:

$$\mathcal{E}(f) := \int_{\mathcal{X} \times \mathbb{R}} (f(x) - y)^2 d\rho(x, y) \quad (3.24)$$

and we learn our function through Mini-batch SGD:

$$\begin{aligned} w_0^q &:= 0, \quad w_{t+1}^q := w_t^q - \mu_t \frac{1}{b} \sum_{i \in B_t} (\langle w_t^q, \phi_M^q(x_i, \gamma_{t,i}) \rangle_{\mathbb{R}^M} - y_i) \phi_M^q(x_i, \gamma_{t,i}) \\ &= (Id - \mu_t C_{M,t}^q) w_t^q + \mu_t \mathbf{b}_t^q \end{aligned} \quad (3.25)$$

where  $b$  is the dimension of the batch, and  $\{B_0, \dots, B_{\mathcal{T}-1}\}$  is the sampling sequence of batches drawn uniformly with replacement from  $\{1, \dots, n\}$ . Where we define:

$$\mathbf{b}_t^q := \frac{1}{b} \sum_{i \in B_t} y_i \phi_M^q(x_i, \gamma_{t,i}) \quad (3.26)$$

and its full-precision analogous as:

$$\mathbf{b}_t := \frac{1}{b} \sum_{i \in B_t} y_i \phi_M(x_i) \quad (3.27)$$

Our estimator function takes the form:

$$f_t^q(\cdot) := \langle w_t^q, \phi_M(\cdot) \rangle_{\mathbb{R}^M}. \quad (3.28)$$

The estimator depends on  $w_t^q$ , which, in turns, depends on the sampling procedure of the quantization noise term  $\gamma$ . To average out the effect of such sampling we consider:

$$\mathbb{E}_\gamma [\mathcal{E}(f_t^q)] = \int_{\Gamma} \int_{\mathcal{X} \times \mathbb{R}} (f_t^q(x; \gamma) - y)^2 d\rho(x, y) d\mathbb{Q}(\gamma) \quad (3.29)$$

we are abusing the notation since we should integrate over  $\Gamma^t$ , but since every drawn is independent we can separate the integral hence keeping a lighter notation.

The aim now is to estimate:

$$\mathbb{E}_\gamma [\mathcal{E}(f_t^q) - \mathcal{E}(f_H)] = \mathbb{E}_\gamma \|f_t^q - f_H\|_\rho^2 \quad (3.30)$$

where the equality is due to a well-known result (see [6]) under assumption (A1) in 3.3. In order to prove a bound on the error we need the following function:

$$f_t := S_M w_t \quad (3.31)$$

where  $w_t$  is learned as in [4]:

$$\begin{aligned} w_0 &:= 0, \quad w_{t+1} := w_t - \frac{\mu_t}{b} \sum_{i \in B_t} (\langle \phi_M((x_i)), w_t \rangle - y_i) \phi_M((x_i)) \\ &= (Id - \mu_t C_{M,t}) w_t + \mu_t \mathbf{b}_t \end{aligned} \quad (3.32)$$

and we use the same convention over the mini-batches.

### 3.3 Bound on the population error of the reference estimator and our stochastic one

To establish the statistical guarantees and convergence rates for the 1-bit stochastically quantized Mini-batch SGD estimator, we formally introduce the following set of assumptions.

**Assumptions:** Let  $\rho$  be the joint probability distribution on  $\mathcal{X} \times \mathcal{Y}$ . We assume that:

- (A1) We have  $f_\rho \in \mathcal{H}$ , where  $f_\rho(x) := \int_{\mathcal{Y}} y d\rho(y|x)$ .
- (A2) There exist  $y_{\max} > 0$  and  $\kappa > 0$  such that  $|y| \leq y_{\max}$   $\rho$ -almost surely; also the feature map is such that  $\|\phi_M(x)\|_{\mathbb{R}^M} \leq \kappa$  for all  $x \in \mathcal{X}$ .
- (A3) For any  $x \in \mathcal{X}$  and  $l \in \mathbb{N}$ ,  $\int_{\mathbb{R}} y^{2l} d\rho(y|x) \leq l! B^l p$  for  $B \in (0, +\infty)$ ,  $p \in (1, +\infty)$ .
- (A4) The step size is constant,  $\mu_t \equiv \mu$ , and satisfies  $\mu (\kappa^2 + \frac{2}{M}) \leq 1$ .
- (A5) There exists  $U < +\infty$ , independent of  $T, M, n$  and  $\mu$ , such that  $\max_{t \leq T} \|u_t\|_{\mathbb{R}^M} \leq U$  almost surely with respect to the sample and to the mini-batch draws, where  $u_t := \mathbb{E}_\gamma[w_t^q | J]$ .
- (A6) Conditionally on  $(\mathcal{X}, \Omega, \xi)$ , for  $j \geq 1$  write  $\hat{A}_j := \hat{C}_{M,j}$  and  $\hat{A}'_j := \hat{C}_{M,j} + \hat{D}_{b,j}$ , and for  $s \geq 1$  set  $\Pi_s := (I - \mu \hat{A}_s) \cdots (I - \mu \hat{A}_1)$ ,  $\Pi'_s := (I - \mu \hat{A}'_s) \cdots (I - \mu \hat{A}'_1)$ ,  $\Pi_0 = \Pi'_0 := I$ , and

$$\mathcal{A}_s := \mathbb{E}_J[\Pi_s^\top C_M \Pi_s], \quad \mathcal{A}'_s := \mathbb{E}_J[(\Pi'_s)^\top C_M \Pi'_s] \quad (3.33)$$

There exists  $c_A \geq 1$ , independent of  $s, T, \mu, M, n$ , such that for every  $0 \leq s < T$

$$\max \{ \|\mathcal{A}_s\|_{op}, \|\mathcal{A}'_s\|_{op} \} \leq c_A \min \{ \kappa^2, \frac{1}{\mu(s \vee 1)} \}.$$

Throughout we write

$$\Theta_T := \mu \sum_{s=0}^{T-1} \|\mathcal{A}_s\|_{op}^{1/2}, \quad \Xi_T := \mu^2 \sum_{s=0}^{T-1} \|\mathcal{A}_s\|_{op}. \quad (3.34)$$

Assumptions (A1)–(A4) are standard in this setting: (A1)–(A3) are Assumptions 3, 2 and 4 of [4], and (A4) is the step-size restriction under which the SGD map is a contraction, with the extra  $2/M$  accounting for the quantization noise. Assumptions (A5) and (A6) are specific to our analysis and are the two we do not prove: (A6) is discussed in Remark 2 and (A5) in Remark 1; removing them is left open.

**Remark 1.** Assumption (A5) is about the scale of  $U$ . Indeed, by (A4) the map  $I - \mu(\hat{C}_{M,t} + \hat{D}_{b,t})$  is a contraction and  $\|\mathbf{b}_t\| \leq \kappa y_{\max}$  by (A2), so (3.41) and  $u_0 = 0$  give  $\|u_t\| \leq \kappa y_{\max} \mu T$  for every  $t \leq T$ : (A5) asks that this bound, which grows like  $\sqrt{n}$  when  $\mu T \asymp \sqrt{n}$ , is improved by a constant.

**Remark 2.** Assumption (A6) is the mini-batch counterpart, in expectation over  $J$ , of an inequality that can be proved in the population case. Indeed, for  $s = 0$  we have  $\Pi_0 = I$ , hence  $\mathcal{A}_0 = C_M$  and  $\|C_M\|_{op} \leq \kappa^2$ , so (A6) holds at  $s = 0$  with  $c_A = 1$ ; and if  $A_j \equiv C_M$  for every  $j$ , then  $\mathcal{A}_s = C_M(I - \mu C_M)^{2s}$  and, since  $\mu\lambda \in [0, 1]$  for every  $\lambda \in \sigma(C_M)$  by (A4),

$$\|\mathcal{A}_s\|_{op} = \max_{\lambda \in [0, \kappa^2]} \lambda (1 - \mu\lambda)^{2s} \leq \frac{1}{\mu(2s+1)},$$

again with  $c_A = 1$ . A proof of (A6) for the mini-batch operators, along the lines of [10], is left as an open problem.

We now state the following, which is the main result that we want to prove:

**Theorem 3.3.1.** Let  $\mathcal{X} \subset \mathbb{R}^d$  be a compact set and let  $\rho$  be a joint probability distribution on  $\mathcal{X} \times \mathcal{Y}$ , with marginal  $\rho_{\mathcal{X}}$ . Assume (A1)–(A6) and let  $f_T^q(\cdot) = \langle w_T^q, \phi_M(\cdot) \rangle_{\mathbb{R}^M}$  be the estimator produced at step  $T$  by 1-bit stochastically quantized mini-batch SGD initialized at  $w_0^q = 0$ . Then, conditionally on  $(\mathcal{Z}, \Omega, \xi)$  we have:

$$\mathbb{E}_{J,\gamma} \left[ \|f_T^q - f_{\mathcal{H}}\|_{L^2(\rho_{\mathcal{X}})}^2 \right] \leq 3\sigma_{\mathcal{H}}^2 \Xi_T + \frac{12U^2}{M^2} \Theta_T^2 + 3\mathbb{E}_J \left[ \|f_T - f_{\mathcal{H}}\|_{L^2(\rho_{\mathcal{X}})}^2 \right], \quad (3.35)$$

where  $f_T(\cdot) = \langle w_T, \phi_M(\cdot) \rangle_{\mathbb{R}^M}$  is the full-precision SGD estimator,  $\Sigma^2 := 4(8\kappa^2 + \frac{64}{M})$ ,  $C_q := 2\Sigma^2 U^2 + 4y_{\max}^2$  and

$$V^2 := \frac{T\mu^2 C_q}{b} \exp\left(\frac{2T\mu^2 \Sigma^2}{b}\right), \quad \sigma_{\mathcal{H}}^2 := \frac{\Sigma^2}{b} (2U^2 + 2V^2) + \frac{4y_{\max}^2}{b}.$$

We have three independent sources of randomness: the sample  $\mathcal{X} = (x_i, y_i)_{i=1}^n$  and the feature draw  $(\Omega, \xi)$ , on which we condition throughout; the mini-batch sequence  $J = \{B_0, \dots, B_{T-1}\}$ , drawn uniformly with replacement; and the noise variables  $\gamma = (\gamma_0, \dots, \gamma_{T-1})$ , redrawn independently at every step. We write  $\mathbb{E}_\gamma[\cdot | J]$  for the expectation over  $\gamma$  at fixed  $J$ , so that  $u_t := \mathbb{E}_\gamma[w_t^q | J]$  is a  $J$ -measurable vector, and  $\mathbb{E}_{J,\gamma}$  for the joint expectation.

*Proof.* Take  $\mathbb{E}_{J,\gamma}$  in the elementary decomposition

$$\|f_T^q - f_{\mathcal{H}}\|_{L^2(\rho_{\mathcal{X}})}^2 \leq 3\|f_T^q - \mathbb{E}_\gamma[f_T^q | J]\|_{L^2(\rho_{\mathcal{X}})}^2 \quad (3.36)$$

$$+ 3\|\mathbb{E}_\gamma[f_T^q | J] - f_T\|_{L^2(\rho_{\mathcal{X}})}^2 \quad (3.37)$$

$$+ 3\|f_T - f_{\mathcal{H}}\|_{L^2(\rho_{\mathcal{X}})}^2. \quad (3.38)$$

The term (3.36) is bounded by Proposition 3.4.6 and the term (3.37), squared, by Proposition 3.4.2.  $\square$

**Remark 3.** Under (A1) we have  $f_{\mathcal{H}} = f_\rho$ , so that  $\mathcal{E}(f) - \mathcal{E}(f_{\mathcal{H}}) = \|f - f_{\mathcal{H}}\|_{L^2(\rho_{\mathcal{X}})}^2$  for every  $f \in L^2(\rho_{\mathcal{X}})$ ; this is what allows us to use the bound for (3.38) described in [4].

**Remark 4.** The third term of (3.35) is the excess risk of the full-precision estimator, bounded in [4, Thm. 1] with probability at least  $1 - \delta$  over  $(\mathcal{X}, \Omega, \xi)$ ; hence in (3.35) analyze the impact of quantization over the original learning procedure.

**Corollary 3.3.2.** Assume (A1)–(A6) and the conditions of Remark 3. Let the free parameters satisfy one of the conditions (c1.1)–(c1.4) of Corollary 1 in [4], under which  $\mu T \asymp \sqrt{n}$  and  $\mathbb{E}_J \|f_T - f_{\mathcal{H}}\|_{L^2(\rho_{\mathcal{X}})}^2 = \mathcal{O}(1/\sqrt{n})$  with probability at least  $1 - \delta$ , and let  $M = \tilde{\mathcal{O}}(\sqrt{n})$ . Then

$$\mathbb{E}_{J,\gamma} [\|f_T^q - f_{\mathcal{H}}\|_{L^2(\rho_{\mathcal{X}})}^2] = \tilde{\mathcal{O}}\left(\frac{1}{\sqrt{n}}\right). \quad (3.39)$$

*Proof.* By (3.40),  $\Theta_T^2 \leq 10c_A \mu T$ , so the bias term of (3.35) is of order  $U^2 c_A \mu T / M^2 \asymp c_A U^2 \sqrt{n} / M^2 = \tilde{\mathcal{O}}(n^{-1/2})$  for  $M = \tilde{\mathcal{O}}(\sqrt{n})$ . For the variance term,  $T\mu^2/b \leq 1$  under each of (c1.1)–(c1.4), hence  $V^2 = \mathcal{O}(1)$  and  $\sigma_{\mathcal{H}}^2 = \mathcal{O}(1/b)$ , while  $\Xi_T = \mathcal{O}(c_A \mu \log T)$ ; the product is  $\tilde{\mathcal{O}}(\mu/b)$ , which is  $\tilde{\mathcal{O}}(n^{-1/2})$  in all four configurations. The third term is  $\mathcal{O}(1/\sqrt{n})$  by assumption.  $\square$

**Remark 5.** The binding term is the bias: the requirement is  $\mu T / M^2 \lesssim n^{-1/2}$ , i.e.  $M \gtrsim \sqrt{n}$  whenever  $\mu T \asymp \sqrt{n}$ . Thus 1-bit stochastic quantization preserves the minimax-optimal rate  $\mathcal{O}(1/\sqrt{n})$  of [3] with the same number of random features as in full precision, and  $M = \tilde{\mathcal{O}}(\sqrt{n})$  is exactly saturating for this bound.

### 3.4 Supporting lemmas for the main result

First we retrieve a convenient bound for operator norm of the iterates.

**Lemma 3.4.1.** Under (A4) and (A6),

$$\Theta_T \leq \sqrt{c_A} \left( \mu \kappa + 2\sqrt{\mu T} \right), \quad \Xi_T \leq c_A \left( \mu^2 \kappa^2 + \mu (1 + \log T) \right).$$

In particular,

$$\Theta_T^2 \leq 10 c_A \mu T, \quad \Xi_T \leq c_A \mu (2 + \log T). \quad (3.40)$$

*Proof.* Since  $\Pi_0 = I$ , we have  $\mathcal{A}_0 = C_M$  and  $\|\mathcal{A}_0\|_{\text{op}} \leq c_A \kappa^2$  by (A6), so the term  $s = 0$  contributes  $\sqrt{c_A} \mu \kappa$  to  $\Theta_T$  and  $c_A \mu^2 \kappa^2$  to  $\Xi_T$ . For  $s \geq 1$  we use the second entry of the minimum in (A6). Since  $\sum_{s=1}^{T-1} s^{-1/2} \leq 2\sqrt{T}$ ,

$$\mu \sum_{s=1}^{T-1} \sqrt{\frac{c_A}{\mu s}} = \sqrt{c_A \mu} \sum_{s=1}^{T-1} s^{-1/2} \leq 2\sqrt{c_A \mu T},$$

and since  $\sum_{s=1}^{T-1} s^{-1} \leq 1 + \log T$ ,

$$\mu^2 \sum_{s=1}^{T-1} \frac{c_A}{\mu s} \leq c_A \mu (1 + \log T).$$

For (3.40), basic inequality gives  $\Theta_T^2 \leq 2c_A (\mu^2 \kappa^2 + 4\mu T)$ ; by (A4),  $\mu \kappa^2 \leq 1$ , hence  $\mu^2 \kappa^2 = \mu \cdot \mu \kappa^2 \leq \mu \leq \mu T$ , and  $\Theta_T^2 \leq 10 c_A \mu T$ . The same substitution gives the bound on  $\Xi_T$ .  $\square$

We start by bounding the second term.

**Proposition 3.4.2.** Let  $f_T^q$  and  $f_T$  be the quantized and full-precision estimators, respectively. Under (A2), (A4), (A5) and (A6),

$$\left( \mathbb{E}_J \left\| \mathbb{E}_\gamma[f_T^q] - f_T \right\|_{L^2(\rho_X)}^2 \right)^{1/2} \leq \frac{2U}{M} \Theta_T.$$

*Proof.* Let  $u_t = \mathbb{E}_\gamma[w_t^q | J]$  as defined in Theorem 3.3.1. Given the mini-batch of indices  $B_k$  at iteration  $k$ , we define

$$\begin{aligned} \hat{C}_{M,k} &:= \frac{1}{b} \sum_{i \in B_k} \phi_M(x_i) \phi_M(x_i)^T \\ \hat{D}_{b,k} &:= \frac{1}{b} \sum_{i \in B_k} D(x_i) \\ \mathbf{b}_k &:= \frac{1}{b} \sum_{i \in B_k} y_i \phi_M(x_i). \end{aligned}$$

Taking  $\mathbb{E}_\gamma$  in (3.25) and using  $\mathbb{E}_{\gamma,i}[\phi_M^q(x_i, \gamma_{t,i})\phi_M^q(x_i, \gamma_{t,i})^T] = \phi_M(x_i)\phi_M(x_i)^T + D(x_i)$  together with  $\mathbb{E}_{\gamma,i}[\phi_M^q(x_i, \gamma_{t,i})] = \phi_M(x_i)$ , the expected trajectory satisfies

$$u_{t+1} = (I - \mu(\hat{C}_{M,t} + \hat{D}_{b,t}))u_t + \mu \mathbf{b}_t, \quad (3.41)$$

while the non-quantized sequence obeys  $w_{t+1} = (I - \mu\hat{C}_{M,t})w_t + \mu \mathbf{b}_t$ . Defining  $e_t := w_t - u_t$  we obtain  $e_{t+1} = (I - \mu\hat{C}_{M,t})e_t + \mu\hat{D}_{b,t}u_t$ , and from  $e_0 = 0$ ,

$$e_T = \mu \sum_{k=0}^{T-1} P_k v_k, \quad P_k := \prod_{j=k+1}^{T-1} (I - \mu\hat{C}_{M,j}), \quad v_k := \hat{D}_{b,k}u_k.$$

Since  $C_M = S_M^* S_M$  we have  $\|S_M v\|_{L^2(\rho_X)} = \|C_M^{1/2} v\|_{\mathbb{R}^M}$ , so by Minkowski's inequality:

$$\left( \mathbb{E}_J \left\| \mathbb{E}_\gamma[f_T^q] - f_T \right\|_{L^2(\rho_X)}^2 \right)^{1/2} = \left( \mathbb{E}_J \|C_M^{1/2} e_T\|^2 \right)^{1/2} \leq \mu \sum_{k=0}^{T-1} \left( \mathbb{E}_J \|C_M^{1/2} P_k v_k\|^2 \right)^{1/2}.$$

The matrix  $P_k$  depends only on the batches  $B_{k+1}, \dots, B_{T-1}$ , whereas  $\hat{D}_{b,k}$  depends on  $B_k$  and  $u_k$  on  $B_0, \dots, B_{k-1}$ ; the batches being i.i.d.,  $P_k$  is independent of  $v_k$  and has the same law as  $\Pi_s$  with  $s := T - 1 - k$ . Therefore, recalling (3.33), we have:

$$\mathbb{E}_J \|C_M^{1/2} P_k v_k\|^2 = \mathbb{E}_J \langle v_k, \mathcal{A}_s v_k \rangle \leq \|\mathcal{A}_s\|_{\text{op}} \mathbb{E}_J \|v_k\|^2 \leq \|\mathcal{A}_s\|_{\text{op}} \left( \frac{2U}{M} \right)^2,$$

where we used that each  $D(x_i)$  is diagonal with  $\|D(x_i)\|_{\text{op}} \leq 2/M$  by Proposition 7 in [20], hence  $\|\hat{D}_{b,k}\|_{\text{op}} \leq \frac{1}{b} \sum_{i \in B_k} \|D(x_i)\|_{\text{op}} \leq \frac{2}{M}$ , together with (A5). Summing over  $k$  and recalling the definition (3.34) of  $\Theta_T$  gives the claim.  $\square$

**Remark 6.** Combining Proposition 3.4.2 with Lemma 3.4.1, the squared tracking bias is of order  $U^2 c_A \mu T / M^2$ : it is the product  $\mu T$ , and not  $\mu$  and  $T$  separately, that governs it. Requiring it to be  $\mathcal{O}(n^{-1/2})$  therefore amounts to  $M^2 \gtrsim \mu T \sqrt{n}$ , which for  $\mu T \asymp \sqrt{n}$  is  $M \gtrsim \sqrt{n}$ .

Now we deal with the variance term:

$$\mathbb{E}_\gamma \left[ \int_{\mathcal{X}} (f_{\mathcal{T}}^q(x) - \mathbb{E}_\gamma(f_{\mathcal{T}}^q)(x))^2 dx \right] \quad (3.42)$$

according to the notation in the previous proof we can rewrite this as:

$$\mathbb{E}_\gamma \left[ \|S_M(w_{\mathcal{T}}^q - u_{\mathcal{T}})\|_{L^2(\mathcal{X})}^2 \right] \quad (3.43)$$

where  $u_t$  is defined as in (3.41). Since  $C_M = S_M^* S_M$ , this equals  $\mathbb{E}_\gamma \left[ \|C_M^{1/2}(w_T^q - u_T)\|_{\mathbb{R}^M}^2 \right]$ . We define:

$$\delta_t := w_t^q - u_t \quad (3.44)$$

recalling (3.41) and working in a similar fashion, we can derive a recursive equation for  $w_t^q$ :

$$w_{t+1}^q = (I - \mu \hat{C}_{M,t}^q) w_t^q + \mu \mathbf{b}_t^q \quad (3.45)$$

hence (3.44) evolves as:

$$\begin{aligned} \delta_{t+1} &= (I - \mu \hat{C}_{M,t}^q) w_t^q + \mu \mathbf{b}_t^q - ((I - \mu(\hat{C}_{M,t} + \hat{D}_{b,t})) u_t + \mu \mathbf{b}_t) \\ &= \delta_t - \mu \hat{C}_{M,t}^q w_t^q + \mu(\hat{C}_{M,t} + \hat{D}_{b,t}) w_t^q - \mu(\hat{C}_{M,t} + \hat{D}_{b,t}) w_t^q + \mu(\hat{C}_{M,t} + \hat{D}_{b,t}) u_t + \mu(\mathbf{b}_t^q - \mathbf{b}_t) \\ &= (I - \mu(\hat{C}_{M,t} + \hat{D}_{b,t})) \delta_t + \mu H_t \end{aligned} \quad (3.46)$$

with:

$$H_t := -(\hat{C}_{M,t}^q - \hat{C}_{M,t} - \hat{D}_{b,t}) w_t^q + (\mathbf{b}_t^q - \mathbf{b}_t) \quad (3.47)$$

so we have:

$$\delta_{t+1}^T \delta_{t+1} = \delta_t^T (I - \mu(\hat{C}_{M,t} + \hat{D}_{b,t}))^2 \delta_t + \mu^2 H_t^T H_t + 2\mu \delta_t^T (I - \mu(\hat{C}_{M,t} + \hat{D}_{b,t})) H_t \quad (3.48)$$

now, both  $w_t^q$  and  $\hat{C}_{M,t} + \hat{D}_{b,t}$  are  $\mathcal{F}_t$ -measurable, while  $\gamma_t$  is independent of  $\mathcal{F}_t$ ; since  $\mathbb{E}_{\gamma_t}[\hat{C}_{M,t}^q] = \hat{C}_{M,t} + \hat{D}_{b,t}$  by (3.22) and  $\mathbb{E}_{\gamma_t}[\mathbf{b}_t^q] = \mathbf{b}_t$ , we have

$$\mathbb{E}_{\gamma_t} [H_t \mid \mathcal{F}_t] = 0. \quad (3.49)$$

The cross term in (3.48) therefore vanishes in conditional expectation, and

$$\mathbb{E}_{J,\gamma} [\|\delta_{t+1}\|^2] = \mathbb{E}_{J,\gamma} [\delta_t^T (I - \mu(\hat{C}_{M,t} + \hat{D}_{b,t}))^2 \delta_t] + \mu^2 \mathbb{E}_{J,\gamma} [\|H_t\|^2]. \quad (3.50)$$

Set now:

$$\Delta_i := \eta_i \phi_i^T + \phi_i \eta_i^T + \eta_i \eta_i^T - D(x_i), \quad (3.51)$$

so that, by the definitions of  $\hat{C}_{M,t}^q$  and  $\mathbf{b}_t^q$ ,

$$H_t = -\frac{1}{b} \sum_{i \in B_t} \Delta_i w_t^q + (\mathbf{b}_t^q - \mathbf{b}_t), \quad \|H_t\|^2 \leq \frac{2}{b^2} \left\| \sum_{i \in B_t} \Delta_i w_t^q \right\|^2 + 2\|\mathbf{b}_t^q - \mathbf{b}_t\|^2. \quad (3.52)$$

The matrices  $(\Delta_i)_{i \in B_t}$  are independent, since the dither is redrawn at every step and independently across the elements of the mini-batch, and centred, since  $\mathbb{E}_{\gamma_t}[\eta_i] = 0$  and  $\mathbb{E}_{\gamma_t}[\eta_i \eta_i^T] = D(x_i)$ ; moreover  $w_t^q$  does not depend on  $\gamma_t$ . The cross terms vanish and therefore:

$$\mathbb{E}_{\gamma_t} \left[ \left\| \sum_{i \in B_t} \Delta_i w_t^q \right\|^2 \right] = \sum_{i \in B_t} \mathbb{E}_{\gamma_t} [\|\Delta_i w_t^q\|^2]. \quad (3.53)$$

Hence we can trivially bound:

$$\mathbb{E}_{\gamma_t} [\|\Delta_i w_t^q\|^2] \leq 2 \mathbb{E}_{\gamma_t} \left\| (\eta_i \phi_i^T + \phi_i \eta_i^T) w_t^q \right\|^2 + 2 \mathbb{E}_{\gamma_t} \left\| (\eta_i \eta_i^T - D(x_i)) w_t^q \right\|^2. \quad (3.54)$$

Now we bound the second term on the l.h.s. of (3.54):

**Lemma 3.4.3.**

$$\mathbb{E}_\gamma \left\| (\eta_i \eta_i^T - \mathbb{E}_\gamma(\eta_i \eta_i^T)) w_t^q \right\|^2 \leq \frac{64}{M} \mathbb{E}_\gamma \|w_t^q\|^2, \quad i \in B_t$$

*Proof.* Let  $x_i$  be a generic sample from the mini-batch at time  $t$ . We denote the  $j$ -th coordinate of  $\eta_i$  as  $\eta_{i,j}$  for  $j \in \{1, \dots, M\}$ . We define  $v_i := (\eta_i \eta_i^T - D(x_i)) w_t^q$ . Its  $j$ -th component is given by:

$$(v_i)_j = (\eta_{i,j}^2 - D(x_i)_{j,j})(w_t^q)_j + \sum_{l \neq j} \eta_{i,j} \eta_{i,l} (w_t^q)_l \quad (3.55)$$

Since the quantization noise coordinates are mutually independent and zero-mean given  $x_i$ , it holds that  $\mathbb{E}[\eta_{i,j} \eta_{i,l}] = 0$  for  $j \neq l$ . Now we take the expectation w.r.t the noise sample at time  $t$ . Recalling that  $w_t^q$  is independent from it we have that the cross terms vanish:

$$\begin{aligned} \mathbb{E}[(v_i)_j^2] &= \mathbb{E}[(\eta_{i,j}^2 - D(x_i)_{j,j})^2] (w_t^q)_j^2 + \sum_{l \neq j} \mathbb{E}[\eta_{i,j}^2 \eta_{i,l}^2] (w_t^q)_l^2 \\ &= \mathbb{E}[(\eta_{i,j}^2 - D(x_i)_{j,j})^2] (w_t^q)_j^2 + \sum_{l \neq j} \mathbb{E}[\eta_{i,j}^2] \mathbb{E}[\eta_{i,l}^2] (w_t^q)_l^2 \end{aligned} \quad (3.56)$$

Recalling (3.18) we have:  $\mathbb{E}[\eta_{i,j}^2] = D(x_i)_{j,j} \leq \frac{2}{M}$ . Since  $\eta_{i,j} = \sqrt{2/M} \bar{\eta}_{i,j}$  with  $|\bar{\eta}_{i,j}| = |\pm 1 - \cos(\omega_j^T x_i + \xi_j)| \leq 2$ , we have  $0 \leq \eta_{i,j}^2 \leq \frac{8}{M}$  and  $0 \leq D(x_i)_{j,j} \leq \frac{2}{M}$ , hence  $|\eta_{i,j}^2 - D(x_i)_{j,j}| \leq \frac{8}{M}$  and  $\mathbb{E}[(\eta_{i,j}^2 - D(x_i)_{j,j})^2] \leq \frac{64}{M^2}$ . For the expectation we have:

$$\mathbb{E}[(v_i)_j^2] \leq \frac{64}{M^2} (w_t^q)_j^2 + \frac{4}{M^2} \sum_{l \neq j} (w_t^q)_l^2 \leq \frac{64}{M^2} \sum_{l=1}^M (w_t^q)_l^2 = \frac{64}{M^2} \|w_t^q\|^2 \quad (3.57)$$

Summing over all  $M$  dimensions and taking the expectation over the full  $\gamma$  process we have:

$$\mathbb{E}_\gamma \left[ \|(\eta_i \eta_i^T - D(x_i)) w_t^q\|^2 \right] = \sum_{j=1}^M \mathbb{E}_\gamma [(v_i)_j^2] \leq \sum_{j=1}^M \frac{64}{M^2} \mathbb{E}_\gamma \|w_t^q\|^2 = \frac{64}{M} \mathbb{E}_\gamma \|w_t^q\|^2 \quad (3.58)$$

□

Now we trivially bound the weighted difference, in norm, of the sampled features. Then, with the next Lemma, we define a bound over the mixed term.

**Lemma 3.4.4.**

$$\mathbb{E}_\gamma \|\mathbf{b}_t^q - \mathbf{b}_t\|^2 \leq \frac{2y_{\max}^2}{b}$$

*Proof.* By definition we have  $\mathbf{b}_t^q = \mathbf{b}_t + \frac{1}{b} \sum_{i \in B_t} \eta_i y_i$

$$\mathbb{E}_\gamma \|\mathbf{b}_t^q - \mathbf{b}_t\|^2 = \frac{1}{b^2} \mathbb{E}_\gamma \left\| \sum_{i \in B_t} \eta_i y_i \right\|^2 \leq \frac{y_{\max}^2}{b^2} \sum_{i \in B_t} \mathbb{E}_\gamma \|\eta_i\|^2 \leq \frac{y_{\max}^2}{b^2} \sum_{i \in B_t} 2 = \frac{2y_{\max}^2}{b}$$

□

**Lemma 3.4.5.**

$$\mathbb{E}_\gamma \left\| (\eta_i \phi_i^T + \phi_i \eta_i^T) w_t^q \right\|^2 \leq 8\kappa^2 \mathbb{E}_\gamma \|w_t^q\|^2, \quad i \in B_t$$

*Proof.* Let  $x_i$  be a generic sample from the mini-batch at time  $t$ . We write  $\phi_i := \phi_M(x_i)$  and  $\eta_i := \eta(x_i, \cdot)$  for brevity. Define  $z_i := (\eta_i \phi_i^T + \phi_i \eta_i^T) w_t^q$ . Its  $j$ -th component is:

$$(z_i)_j = \eta_{i,j} \langle \phi_i, w_t^q \rangle_{\mathbb{R}^M} + \phi_{i,j} \langle \eta_i, w_t^q \rangle_{\mathbb{R}^M} \quad (3.59)$$

Expanding the second inner product, we can write:

$$(z_i)_j = \eta_{i,j} \langle \phi_i, w_t^q \rangle + \phi_{i,j} \sum_{l=1}^M \eta_{i,l} (w_t^q)_l \quad (3.60)$$

We compute  $\mathbb{E}_\gamma [(z_i)_j^2]$ , recalling that the quantization noise coordinates  $\eta_{i,1}, \dots, \eta_{i,M}$  are mutually independent and zero-mean given  $x_i$ , and that  $w_t^q$  is independent of  $\gamma$ . Expanding the square:

$$(z_i)_j^2 = \eta_{i,j}^2 \langle \phi_i, w_t^q \rangle^2 + \phi_{i,j}^2 \left( \sum_{l=1}^M \eta_{i,l} (w_t^q)_l \right)^2 + 2\eta_{i,j} \phi_{i,j} \langle \phi_i, w_t^q \rangle \sum_{l=1}^M \eta_{i,l} (w_t^q)_l \quad (3.61)$$

We take the expectation w.r.t.  $\gamma$  term by term.

$$\mathbb{E}_\gamma [\eta_{i,j}^2 \langle \phi_i, w_t^q \rangle^2] = \mathbb{E}_\gamma [\eta_{i,j}^2] \langle \phi_i, w_t^q \rangle^2 = D(x_i)_{j,j} \langle \phi_i, w_t^q \rangle^2 \quad (3.62)$$

where  $\langle \phi_i, w_t^q \rangle$  does not depend on  $\gamma$ .

By independence of the noise coordinates:

$$\mathbb{E}_\gamma \left[ \left( \sum_l \eta_{i,l} (w_t^q)_l \right)^2 \right] = \sum_{l=1}^M \mathbb{E}_\gamma [\eta_{i,l}^2] (w_t^q)_l^2 = \sum_{l=1}^M D(x_i)_{l,l} (w_t^q)_l^2 \quad (3.63)$$

Isolating the  $l = j$  contribution in the sum:

$$\mathbb{E}_\gamma \left[ \eta_{i,j} \sum_l \eta_{i,l} (w_t^q)_l \right] = \mathbb{E}_\gamma [\eta_{i,j}^2] (w_t^q)_j = D(x_i)_{j,j} (w_t^q)_j \quad (3.64)$$

since  $\mathbb{E}[\eta_{i,j} \eta_{i,l}] = 0$  for  $l \neq j$ .

Putting together, using the bound  $D(x_i)_{j,j} \leq \frac{2}{M}$ :

$$\begin{aligned} \mathbb{E}_{\mathcal{H}}[(z_i)_j^2] &= D(x_i)_{j,j} \langle \phi_i, w_t^q \rangle^2 + \phi_{i,j}^2 \sum_l D(x_i)_{l,l} (w_t^q)_l^2 + 2\phi_{i,j} D(x_i)_{j,j} (w_t^q)_j \langle \phi_i, w_t^q \rangle \\ &\leq \frac{2}{M} \langle \phi_i, w_t^q \rangle^2 + \frac{2}{M} \phi_{i,j}^2 \|w_t^q\|^2 + 2\phi_{i,j} D(x_i)_{j,j} (w_t^q)_j \langle \phi_i, w_t^q \rangle \end{aligned} \quad (3.65)$$

Now we sum over  $j = 1, \dots, M$ . For the first term:

$$\sum_{j=1}^M \frac{2}{M} \langle \phi_i, w_t^q \rangle^2 = 2 \langle \phi_i, w_t^q \rangle^2 \leq 2 \|\phi_i\|^2 \|w_t^q\|^2 \leq 2\kappa^2 \|w_t^q\|^2$$

For the second term:

$$\sum_{j=1}^M \frac{2}{M} \phi_{i,j}^2 \|w_t^q\|^2 = \frac{2}{M} \|\phi_i\|^2 \|w_t^q\|^2 \leq \frac{2\kappa^2}{M} \|w_t^q\|^2$$

For the third term, using  $|D(x_i)_{j,j}| \leq \frac{2}{M}$ :

$$\begin{aligned} \left| \sum_{j=1}^M 2\phi_{i,j} D(x_i)_{j,j} (w_t^q)_j \langle \phi_i, w_t^q \rangle \right| &\leq \frac{4}{M} |\langle \phi_i, w_t^q \rangle| \sum_j |\phi_{i,j}| |(w_t^q)_j| \\ &\leq \frac{4}{M} \|\phi_i\| \|w_t^q\| \cdot \|\phi_i\| \|w_t^q\| \\ &= \frac{4\kappa^2}{M} \|w_t^q\|^2 \end{aligned}$$

Combining all three contributions:

$$\begin{aligned} \sum_{j=1}^M \mathbb{E}_{\mathcal{H}}[(z_i)_j^2] &\leq 2\kappa^2 \|w_t^q\|^2 + \frac{2\kappa^2}{M} \|w_t^q\|^2 + \frac{4\kappa^2}{M} \|w_t^q\|^2 \\ &= \kappa^2 \left( 2 + \frac{6}{M} \right) \|w_t^q\|^2 \leq 8\kappa^2 \|w_t^q\|^2. \end{aligned} \quad (3.66)$$

Taking the full expectation over  $\gamma$  yields:

$$\mathbb{E}_{\gamma} [\|(\eta_i \phi_i^T + \phi_i \eta_i^T) w_t^q\|^2] \leq 8\kappa^2 \mathbb{E}_{\gamma} \|w_t^q\|^2 \quad (3.67)$$

□

By Lemmas 3.4.5 and 3.4.3, each summand on the right-hand side of (3.54) is bounded by  $2 \left( 8\kappa^2 + \frac{64}{M} \right) \|w_t^q\|^2$ , uniformly in  $i$ . The orthogonality (3.53) turns the first

summand of (3.52) into a sum of  $b$  such terms, while its second summand,  $2\|\mathbf{b}_t^q - \mathbf{b}_t\|^2$ , is bounded by Lemma 3.4.4. Hence

$$\begin{aligned}\mathbb{E}_\gamma [\|H_t\|^2] &\leq \frac{2}{b^2} \cdot b \cdot 2 \left( 8\kappa^2 + \frac{64}{M} \right) \|w_t^q\|^2 + 2 \cdot \frac{2y_{\max}^2}{b} \\ &= \frac{\Sigma^2}{b} \|w_t^q\|^2 + \frac{4y_{\max}^2}{b}, \quad \Sigma^2 := 4 \left( 8\kappa^2 + \frac{64}{M} \right).\end{aligned}\quad (3.68)$$

This last preposition serves as bound for the variance term. It puts together the effects of the quadratic term, the linear term and the difference of the biased term with respect to the exact one.

**Proposition 3.4.6** (Variance bound on  $f_T^q$ ). Under (A1)–(A6), for all  $T \geq 1$  we have:

$$\mathbb{E}_{J,\gamma} \left[ \|f_T^q - \mathbb{E}_\gamma(f_T^q)\|_{L^2(\rho_X)}^2 \right] \leq \sigma_H^2 \Xi_T,$$

where  $\Sigma^2 := 4(8\kappa^2 + \frac{64}{M})$ ,  $C_q := 2\Sigma^2 U^2 + 4y_{\max}^2$  and

$$V^2 := \frac{T\mu^2 C_q}{b} \exp\left(\frac{2T\mu^2 \Sigma^2}{b}\right), \quad \sigma_H^2 := \frac{\Sigma^2}{b} (2U^2 + 2V^2) + \frac{4y_{\max}^2}{b}.$$

*Proof.* Throughout the proof  $\mathbb{E}_{J,\gamma}$  denotes the expectation over the mini-batch sequence  $J$  and the noise variable  $\gamma$ , conditionally on  $(\mathcal{Z}, \Omega, \xi)$ ; conditional expectations  $\mathbb{E}[\cdot | \mathcal{F}_t]$  carry no subscript, the conditioning already identifying the remaining randomness. Since  $f_T^q = S_M w_T^q$ ,  $\mathbb{E}_\gamma[f_T^q | J] = S_M u_T$  and  $\|S_M v\|_{L^2(\rho_X)} = \|C_M^{1/2} v\|_{\mathbb{R}^M}$ , the claim is equivalent to  $\mathbb{E}_{J,\gamma} \|C_M^{1/2} \delta_T\|^2 \leq \sigma_H^2 \Xi_T$ .

We start by showing:  $\sup_{t \leq T} \mathbb{E}_{J,\gamma} \|\delta_t\|^2 \leq V^2$ . Let  $\mathcal{F}_t$  be the filtration generated by the noise variables  $\gamma_0, \dots, \gamma_{t-1}$  and by the whole mini-batch sampling sequence  $B_0, \dots, B_{T-1}$ . By (A4) the eigenvalues of  $\mu(\hat{C}_{M,t} + \hat{D}_{b,t})$  lie in  $[0, 1]$ , hence  $\|I - \mu(\hat{C}_{M,t} + \hat{D}_{b,t})\|_{\text{op}} \leq 1$ ; the cross term in (3.46) vanishes because  $\mathbb{E}[H_t | \mathcal{F}_t] = 0$ , so

$$\mathbb{E}_{J,\gamma} \|\delta_{t+1}\|^2 \leq \mathbb{E}_{J,\gamma} \|\delta_t\|^2 + \mu^2 \mathbb{E}_{J,\gamma} \|H_t\|^2.$$

By (3.68),  $\mathbb{E}[\|H_t\|^2 | \mathcal{F}_t] \leq \frac{\Sigma^2}{b} \|w_t^q\|^2 + \frac{4y_{\max}^2}{b}$ , and  $\|w_t^q\|^2 \leq 2\|u_t\|^2 + 2\|\delta_t\|^2 \leq 2U^2 + 2\|\delta_t\|^2$  by (A5), hence

$$\mathbb{E}_{J,\gamma} \|\delta_{t+1}\|^2 \leq \left( 1 + \frac{2\mu^2 \Sigma^2}{b} \right) \mathbb{E}_{J,\gamma} \|\delta_t\|^2 + \frac{\mu^2 C_q}{b}.$$

Setting  $x := 2\mu^2 \Sigma^2 / b$  and unrolling from  $\delta_0 = 0$ ,

$$\mathbb{E}_{J,\gamma} \|\delta_T\|^2 \leq \frac{\mu^2 C_q}{b} \sum_{k=0}^{T-1} (1+x)^k = \frac{C_q}{2\Sigma^2} [(1+x)^T - 1] \leq \frac{C_q}{2\Sigma^2} T x e^{Tx} = \frac{T\mu^2 C_q}{b} e^{Tx} = V^2,$$

using  $(1+x)^T \leq e^{Tx}$  and  $e^z - 1 \leq ze^z$  for  $z \geq 0$ . The right-hand side is increasing in  $T$ , so the bound holds for every  $t \leq T$ ; consequently  $\mathbb{E}_{J,\gamma} \|H_t\|^2 \leq \sigma_H^2$  for every  $t < T$ .

Now we want to bound the recursion generating  $\delta_T$ . We unroll (3.46):

$$\delta_T = \mu \sum_{k=0}^{T-1} P'_k H_k, \quad P'_k := \prod_{j=k+1}^{T-1} (I - \mu(\hat{C}_{M,j} + \hat{D}_{b,j})), \quad P'_{T-1} = I.$$

For  $k < l$  the vectors  $C_M^{1/2} P'_k H_k$  and  $C_M^{1/2} P'_l H_l$  are orthogonal in expectation. Indeed  $P'_k$  and  $P'_l$  depend only on the mini-batches, and  $H_k$  on  $B_0, \dots, B_k$  and  $\gamma_0, \dots, \gamma_k$  with  $k \leq l-1$ , so  $(P'_l)^\top C_M P'_k H_k$  is  $\mathcal{F}_l$ -measurable and, by the tower property and by (3.49) we have:

$$\begin{aligned} \mathbb{E}_{J,\gamma} \langle C_M^{1/2} P'_k H_k, C_M^{1/2} P'_l H_l \rangle &= \mathbb{E}_{J,\gamma} \left[ \mathbb{E} [\langle (P'_l)^\top C_M P'_k H_k, H_l \rangle \mid \mathcal{F}_l] \right] \\ &= \mathbb{E}_{J,\gamma} \langle (P'_l)^\top C_M P'_k H_k, \mathbb{E}[H_l \mid \mathcal{F}_l] \rangle = 0. \end{aligned}$$

Furthermore  $P_k$  depends only on  $B_{k+1}, \dots, B_{T-1}$  while  $H_k$  depends on  $B_0, \dots, B_k$  and  $\gamma_0, \dots, \gamma_k$ , so  $P_k$  is independent of  $H_k$  and has the same law as  $\Pi'_s$ ,  $s := T-1-k$ ; hence, by the definition (3.33) of  $\mathcal{A}'_s$ ,

$$\mathbb{E}_{J,\gamma} \|C_M^{1/2} P'_k H_k\|^2 = \mathbb{E}_{J,\gamma} \langle H_k, \mathcal{A}'_s H_k \rangle \leq \|\mathcal{A}'_s\|_{op} \mathbb{E}_{J,\gamma} \|H_k\|^2 \leq \|\mathcal{A}'_s\|_{op} \sigma_{\mathcal{H}}^2.$$

Expanding  $\|C_M^{1/2} \delta_T\|^2$  and discarding the cross terms, which vanish by the display above, gives an exact orthogonal decomposition, so that

$$\mathbb{E}_{J,\gamma} \|C_M^{1/2} \delta_T\|^2 = \mu^2 \sum_{k=0}^{T-1} \mathbb{E}_{J,\gamma} \|C_M^{1/2} P'_k H_k\|^2 \leq \sigma_{\mathcal{H}}^2 \mu^2 \sum_{s=0}^{T-1} \|\mathcal{A}'_s\|_{op} = \sigma_{\mathcal{H}}^2 \Xi_T,$$

by the definition (3.34) of  $\Xi_T$ . The only properties used are  $\mathbb{E}[H_t \mid \mathcal{F}_t] = 0$ , (A6) and the uniform bound  $\mathbb{E}_{J,\gamma} \|H_k\|^2 \leq \sigma_{\mathcal{H}}^2$  established in the first step.  $\square$

## Experiments

Throughout this chapter  $b$  denotes the mini-batch size, as in Chapter 3, and the quantizer is always the 1-bit stochastic rounding of Section 1.2.2: every coordinate of  $\phi_M(x)$  is mapped to a vertex of  $\{-\sqrt{2/M}, +\sqrt{2/M}\}^M$  according to (4.2) below.

### 4.1 Kernel Approximation Quality

Before presenting the learning performance we show how the two different quantized approximations, stochastic and deterministic, approximate the kernel function on pairs of points taken from the CASP dataset. Figures 4.1–4.2 plot  $\hat{k}_{M,q}(x_1, x_2)$  against  $\|x_1 - x_2\|$ . Deterministic (sign) quantization is visibly biased near the origin: it estimates the scale mixture  $\kappa_{q,q}^\Delta(u) = \sum_{h \text{ odd}} |S_h|^2 \kappa^\Delta(hu)$  of [16] rather than  $\kappa^\Delta$  itself, and the distortion is a deterministic feature of the map, independent of the value of  $M$ .

Stochastic quantization is unbiased under the condition of the independence of the noise variables: writing  $\hat{k}_{M,q}(x_1, x_2) = \langle \phi_M^q(x_1, \gamma), \phi_M^q(x_2, \gamma') \rangle$ , the estimator satisfies  $\mathbb{E}[\hat{k}_{M,q}(x_1, x_2)] = k(x_1, x_2)$  whenever the two noise variables  $\gamma, \gamma'$  are drawn independently, and this holds *also* when  $x_1 = x_2$ . What breaks unbiasedness is reusing the same realisation of the quantized feature for both arguments, i.e.  $\gamma = \gamma'$ : in that case each coordinate contributes  $(\pm R)^2 = 2/M$  deterministically, so  $\hat{k}_{M,q}(x, x) = \|\phi_M^q(x, \gamma)\|^2 = 2$  instead of  $k(x, x) = 1$ . The diagonal of the plot is therefore the only place where the estimate is biased, and the bias comes from the coupling of the two features, independently of the points coinciding.

As we will see, retrieving in expectation the starting kernel is neither a sufficient nor a necessary condition for being a good estimator in our learning procedure. In fact, in some cases, the estimator retrieved by the deterministic quantization procedure is better than the stochastic one.

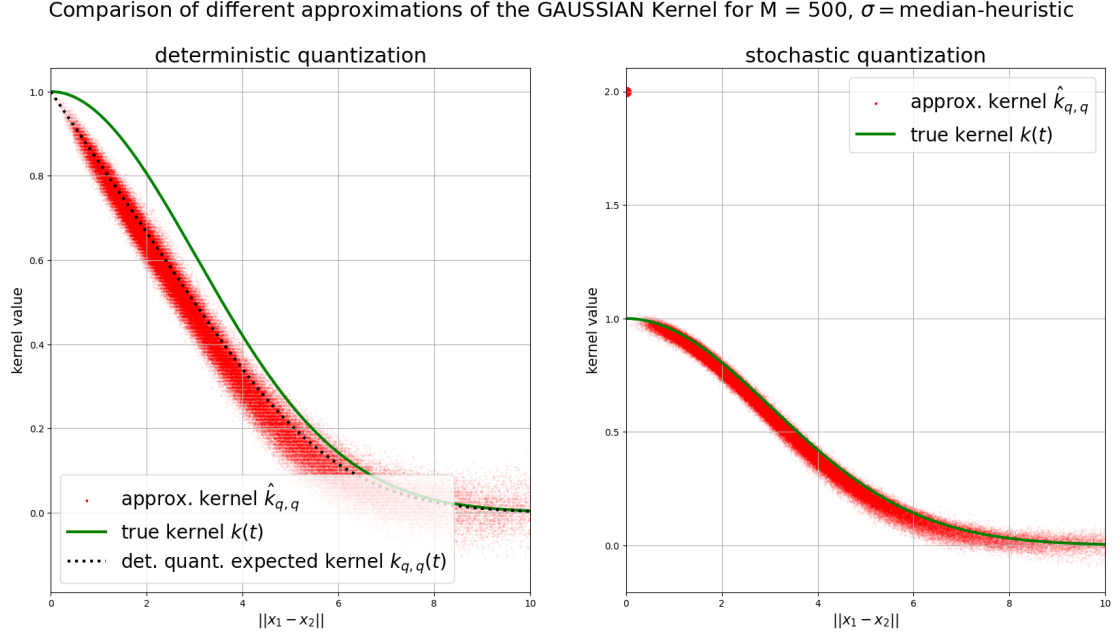


Figure 4.1: We plot the recovered kernel compared with the original one. In particular we consider  $k$  as the standard Gaussian kernel with  $\sigma$  equal to the median heuristic. On the left, as analyzed in [16], we don't recover the original Gaussian kernel but a biased version of it as in (1.5).

## 4.2 The estimators

We use the map of Definition 1.2.11:

$$\phi_M(x) = \sqrt{\frac{2}{M}} \cos(\Omega^\top x + \xi), \quad \omega_j \stackrel{\text{iid}}{\sim} \Lambda, \quad \xi_j \stackrel{\text{iid}}{\sim} \text{Unif}[0, 2\pi], \quad (4.1)$$

with  $\Lambda = \mathcal{N}(0, \sigma^{-2}I_d)$  for the Gaussian kernel and the multivariate Student- $t$  with  $2\nu$  degrees of freedom for the Matérn kernel<sup>1</sup> of smoothness  $\nu$  and lengthscale  $\rho$ . Writing  $c_j(x) := \cos(\omega_j^\top x + \xi_j)$  and  $R := \sqrt{2/M}$ , the one-bit stochastic quantizer of Section 1.2.2 reads

$$\phi_M^q(x, \gamma)_j = \begin{cases} +R & \text{w.p. } \frac{1}{2}(1 + c_j(x)), \\ -R & \text{w.p. } \frac{1}{2}(1 - c_j(x)), \end{cases} \quad (4.2)$$

and the deterministic one is  $\phi_M^{\text{det}}(x) = M^{-1/2} \text{sgn}(\cos(\Omega^\top x + \xi))$ , both normalised to a constant norm,  $\|\phi_M^q\|^2 \equiv 2$  and  $\|\phi_M^{\text{det}}\|^2 \equiv 1$ .

<sup>1</sup>Being  $d = \|x_1 - x_2\|_2$ , the Matérn kernel is defined as  $C_\nu(d) := \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left( \sqrt{2\nu} \frac{d}{\rho} \right)^\nu K_\nu(\sqrt{2\nu} \frac{d}{\rho})$ , where  $\Gamma$  is the Gamma function,  $K_\nu$  the modified Bessel function of the second kind and  $\rho, \nu > 0$ .

Comparison of different approximations of the MATERN kernel for  $M = 500$ ,  $\rho = \text{median heuristic}$ ,  $\nu = 0.5$

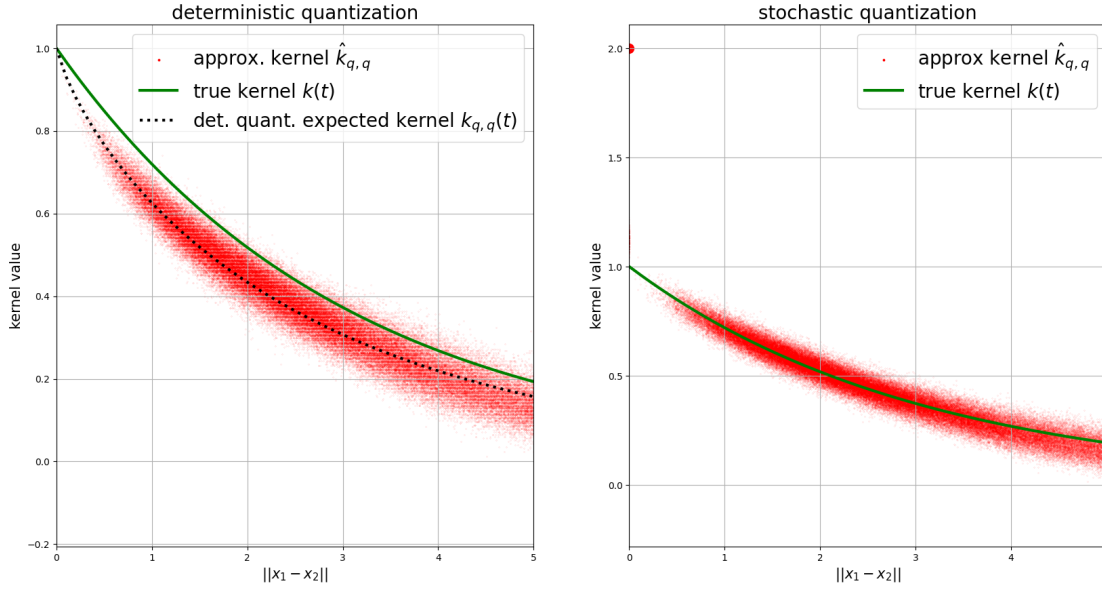


Figure 4.2: Approximation quality of the Matérn kernel, with  $\nu = 1/2$  and  $\rho$  equal to the median heuristic, i.e. the median of the pairwise distances of the points. In both this comparison and the one in Figure 4.1 the stochastic estimate is unbiased away from the diagonal; on the diagonal the two arguments share the same noise variable, and each coordinate contributes  $2/M$  deterministically. Deterministic quantization is biased everywhere except at the origin.

All estimators are trained by the recursion (3.25),

$$w_0 = 0, \quad w_{t+1} = w_t - \frac{\mu_t}{b} \sum_{i \in B_t} \left( \langle w_t, \psi_i \rangle_{\mathbb{R}^M} - y_i \right) \psi_i \quad (4.3)$$

and differ only in the features  $\psi_i$ .

Except for DQ the estimator is  $f_T(\cdot) = \langle w_T, \phi_M(\cdot) \rangle_{\mathbb{R}^M}$  as in [4], i.e. evaluated on full-precision features, since that is the object Chapter 3 bounds. DQ is instead evaluated on its own features: by (1.5) it converges to  $\sum_{h \text{ odd}} |S_h|^2 k^\Delta(h(x-y))$  with  $S_h = \frac{2}{\pi h} (-1)^{(h-1)/2}$ , a linear combination of kernels and therefore a kernel, whose RKHS is larger, as discussed in Section 1.2.2, than the original one [6]. Depending on whether the target is better approximated by the mixture kernel than by  $k^\Delta$  itself, DQ may perform better than the stochastic estimators, and testing it on  $\phi_M$  would vanish that enlarged approximation capability.

| name | features $\psi_i$                                       | stored bits | cost per step |
|------|---------------------------------------------------------|-------------|---------------|
| FP   | $\phi_M(x_i)$ , recomputed                              | —           | $O(bMd)$      |
| SQ   | $\phi_M^q(x_i, \gamma_i)$ , sampled $\gamma_i$          | —           | $O(bMd)$      |
| FQ   | $\phi_M^q(x_i, \gamma_i)$ , $\gamma_i$ fixed at $t = 0$ | $nM$        | $O(bM)$       |
| DQ   | $\phi_M^{\text{det}}(x_i)$                              | $nM$        | $O(bM)$       |

Table 4.1: The four estimators. FP is the full-precision recursion and SQ the quantized recursion analysed in Chapter 3, which redraws the noise variable at every visit of a sample. FQ quantizes once and stores one bit per coordinate; the two would coincide in law if the  $B_t$  were drawn without replacement with  $Tb \leq n$ , so that each sample is used at most once; under the sampling scheme of [4] this fails, and Section 4.3.2 measures the difference.

### 4.3 Configurations and results

**Data** The datasets are CASP ( $d = 9$ , target the RMSD of the tertiary structure, OpenML 42903) and SUPERCONDUCT ( $d = 81$ , target the critical temperature, OpenML superconduct v1), each subsampled to  $10^4$  points and split 80/20, so that  $n = 8000$  and  $n_{\text{test}} = 2000$ . Inputs are standardised coordinate-wise and targets are centred and scaled to unit variance. The bandwidth  $\sigma$  and the lengthscale  $\rho$  are set to the median of the pairwise distances, the median heuristic analysed in [7]. A one-dimensional synthetic problem, on which the target function is known exactly, is used for the trajectory plots of Figures 4.3 and 4.4.

**Shared randomness.** At each value of  $M$  the four estimators of Table 4.1 share the same draw of  $(\Omega, \xi)$ , the same dither seed and the same sequence of mini-batch indices, so that the differences between the curves are attributable to the quantization alone, and not to the random feature draw or to the sampling order.

**Schedules.** Two parameter choices are used, both admissible in Corollary 3.3.2, with  $b = 1$  and a constant step size:

| regime      | $T$         | $\mu$      | $\mu T$    | passes     |
|-------------|-------------|------------|------------|------------|
| single-pass | $n$         | $n^{-1/2}$ | $\sqrt{n}$ | 1          |
| multi-pass  | $n\sqrt{n}$ | $n^{-1}$   | $\sqrt{n}$ | $\sqrt{n}$ |

Table 4.2: Schedule parameters.

In both regimes, for the SQ estimator, the mini-batches are drawn with replacement and are therefore independent, as required by (A6), and SQ redraws its noise variable at every step, so Theorem 3.3.1 applies to SQ in both. They share the same  $\mu T$ , hence the same implicit regularization  $\lambda = 1/(\mu T) = n^{-1/2}$ , and differ only in how many times

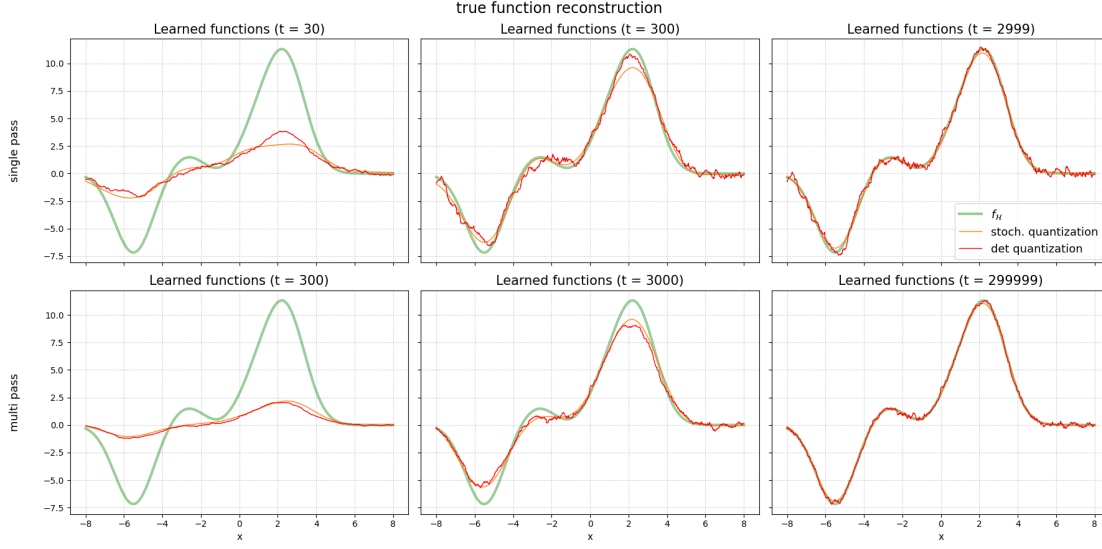


Figure 4.3: Reconstruction of a one-dimensional target function at three stages of training. Top row: single-pass regime ( $t = 300, 3000, 29999$ ); bottom row: multi-epoch regime ( $t = 300, 3000, 29999$ ). The estimators are compared with the target  $f_{\mathcal{H}}$ .

each sample is visited: once on average in the first,  $\sqrt{n}$  times in the second. This is what makes them the right pair for Section 4.3.2, since FQ reuses its noise variable exactly as often as the sample is revisited. The stability condition of (A4) is satisfied throughout: with  $\|\phi_M^q\|^2 \equiv 2$  we have  $\kappa^2 = 2$  and  $\mu(\kappa^2 + 2/M) \leq 0.025$ . In Section 4.3.2, the frozen estimator FQ will break both regimes, while the standard one, SQ, will remain coherent with them hence with the previous analysis. For the single-pass we will train the frozen estimator on a permutation of the training data, hence losing the independence of the batch sampling. For the multi-pass regime we will keep fixed the quantization noise for all the epochs, hence keeping batch independence but losing quantization independence. Experiments suggest that both hypothesis can be relaxed, since the performance of the two estimators are comparable.

### 4.3.1 Validation of the theory

We begin with the trajectory itself. Figure 4.3 shows that the quantized estimator recovers the target on a one-dimensional problem at the same pace as full precision, and Figure 4.4 that the excess risk settles on the reference  $n^{-1/2}$ . This is the rate of Corollary 3.3.2, and the first thing the theory predicts: under our assumptions, quantization does not change the optimal rate of convergence [10]. The two regimes are the one described in Table 4.2.

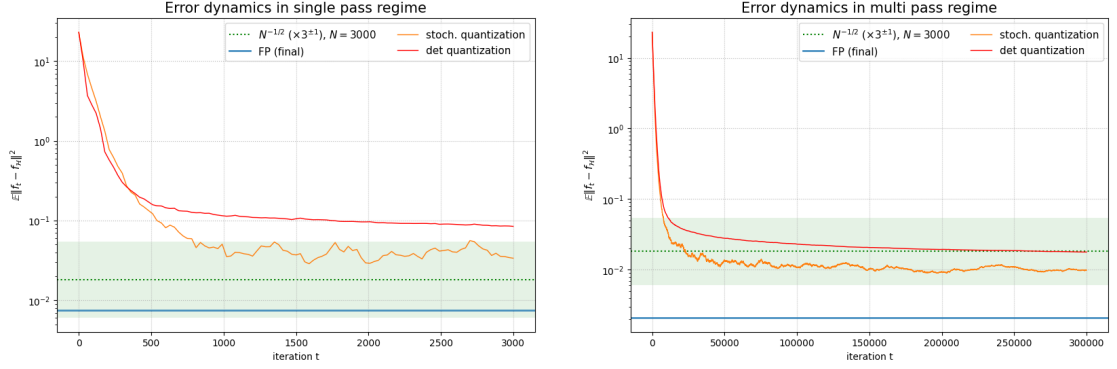


Figure 4.4: Excess risk  $\|f_t - f_H\|_p^2$  along the SGD trajectory, in the single-pass (left,  $n = 3000$ ) and multi-pass (right,  $n = 3000$ ) regimes. The dotted line is the reference rate  $\propto n^{-1/2}$  and the shaded band its tolerance.

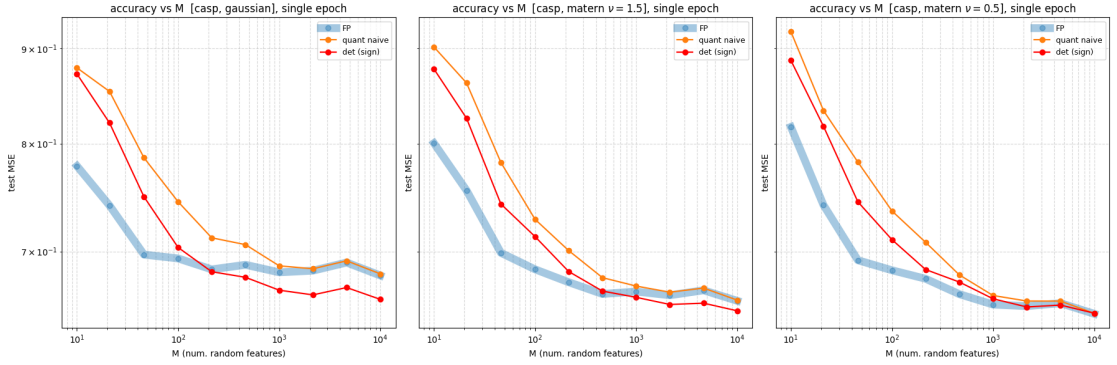


Figure 4.5: Test MSE against the number of random features  $M$  on CASP, in the single-pass schedule of Section 4.3, which is the one covered by Theorem 3.3.1: Gaussian kernel (left), Matérn kernel with  $\nu = 3/2$  (centre) and with  $\nu = 1/2$  (right). FP is the full-precision recursion, *quant naive* the streaming quantized one SQ of (4.2), *det (sign)* the deterministic one DQ, evaluated on its own features. SQ is above FP when the features are few and the two become indistinguishable as  $M$  grows.

**Test error against the number of features.** The second prediction concerns  $M$ . Theorem 3.3.1 adds to the full-precision bound two terms that vanish as the features grow, so the quantized estimators should be worse than full precision when the features are few and should become indistinguishable from it once  $M$  is large enough.

The grid is  $M \in \{10, \dots, 10^4\}$ , ten points logarithmically spaced, and six runs are reported: CASP with the Matérn kernel of smoothness  $\nu \in \{1/2, 3/2\}$  and with the Gaussian kernel in Figure 4.5, SUPERCONDUCT with the same kernels in Figure 4.6.

In every configuration the stochastically quantized estimators are worse than full precision in the low-feature regime and the gap closes monotonically as  $M$  grows: at

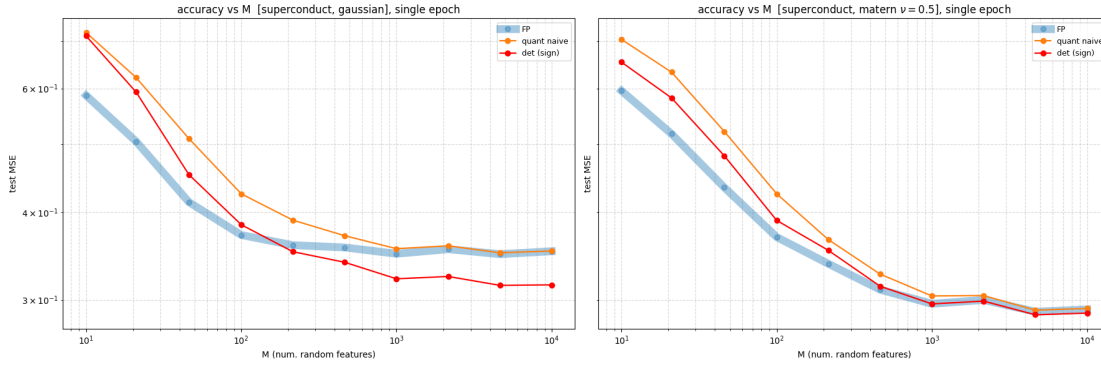


Figure 4.6: Test MSE against  $M$  on SUPERCONDUCT, with the same schedule and the same estimators as in Figure 4.5: Gaussian kernel (left) and Matérn kernel with  $\nu = 1/2$  (right).

$M = 10$  their excess is of the order of ten per cent of the full-precision error; on CASP the curves enter the full-precision band for  $M \gtrsim 10^3$ , on SUPERCONDUCT a small gap survives up to  $M \simeq 10^3$  and disappears at the largest values of the grid. The statistical price of quantization is therefore paid only when the features are too few, which is what the bound says. All the curves flatten at large  $M$  on a baseline set by the noise level of the data and by the finite optimization horizon, so that adding features beyond that point does not improve any of the estimators.

**Deterministic quantization.** The deterministic estimator DQ, which the analysis does not cover, is reported in the same figures and behaves differently from the stochastic ones. It lies strictly between full precision and the stochastic pair for small and moderate  $M$ , it crosses the full-precision curve at a few hundred features, and at the largest values of  $M$  it is at least as good as full precision in every run. The margin depends on the kernel: it is clear with the Gaussian kernel on both datasets, while with the Matérn kernel of smoothness  $\nu = 1/2$  the two are indistinguishable at the right end of the grid. This is consistent with the fact that DQ is evaluated on its own features and therefore targets the mixture kernel of (1.5) rather than  $k^\Delta$ : its RKHS is larger, and on these datasets the additional capacity outweighs the distortion of the recovered kernel documented in Section 4.1, once enough features are available to exploit it. Reproducing the target kernel in expectation is therefore neither necessary nor sufficient for good learning performance. What the present analysis establishes is that at fixed model one bit costs nothing, in neither rate nor features, so that the advantage of DQ here is attributable to the kernel it targets and not to the bit depth — which is what makes it the first step towards an analysis of DQ rather than an endpoint. Compare Figures 4.1–4.2 with Figures 4.7–4.8.

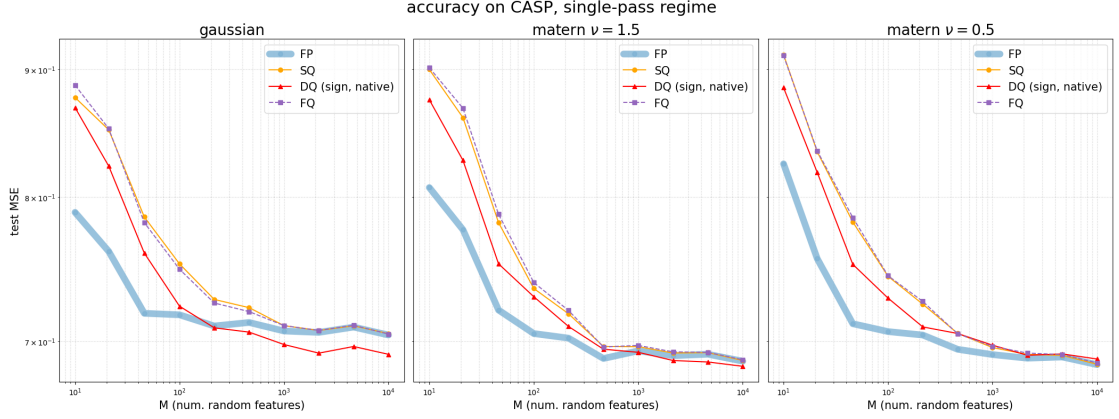


Figure 4.7: Test MSE against  $M$  on CASP in the constant-step schedule: Gaussian kernel (left), Matérn kernel with  $\nu = 3/2$  (centre) and with  $\nu = 1/2$  (right). SQ and DQ are run with mini-batches drawn with replacement, FQ on a single pass over a random permutation, at the same  $\mu$ ,  $T$  and  $b$ ; in both runs every noise variable is used exactly once, so the two quantized curves differ only in the sampling scheme. The band is the full-precision recursion under sampling with replacement; the one run on the permutation agrees with it to within 0.002 and is not shown.

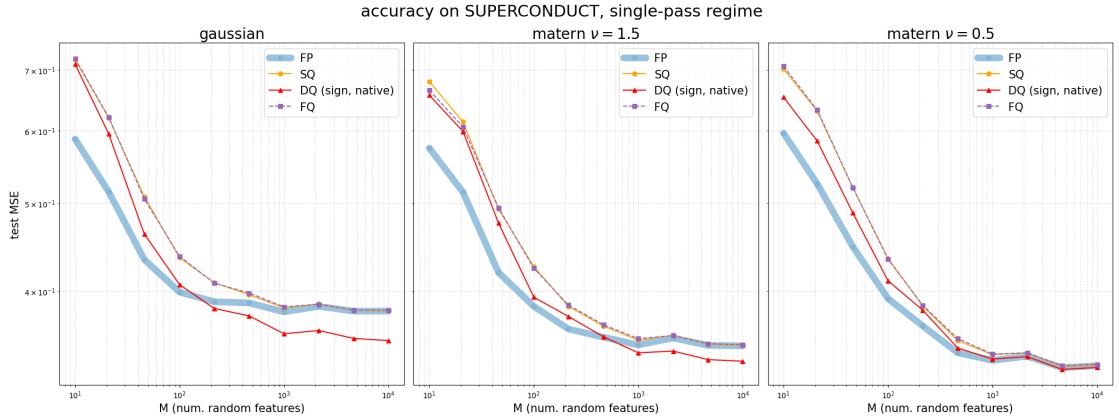


Figure 4.8: Test MSE against  $M$  on SUPERCONDUCT in the single-pass schedule, same kernels and same estimators as in Figure 4.7.

### 4.3.2 Streaming versus frozen

The estimator analysed in Chapter 3 is SQ, which redraws the noise variable at every visit and stores no representation. The estimator that actually keeps one bit per coordinate is FQ, and it is covered in no parameter configuration, as recalled in Section 1.3 and Table 4.2. This subsection compares the two empirically.

Two groups of runs are reported. The first, in Figures 4.7 and 4.8, covers both

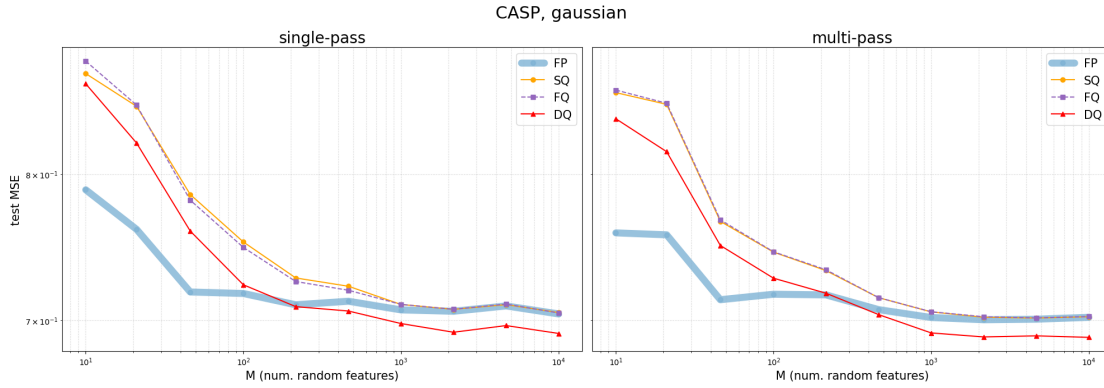


Figure 4.9: Test MSE as a function of the number of random features  $M$  on CASP with the Gaussian kernel, in the single-pass (left) and multi-epoch (right) regimes. The SQ and FQ curves overlap in both panels.

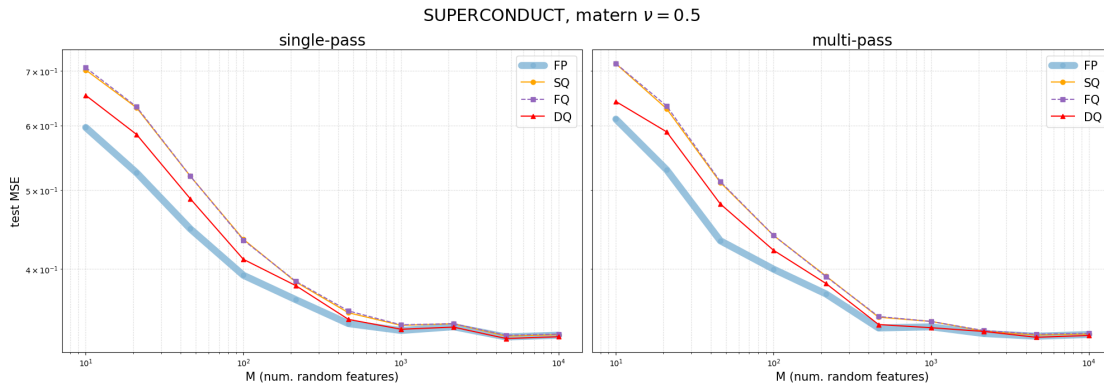


Figure 4.10: Test MSE versus  $M$  on SUPERCONDUCT with the Matérn kernel ( $\nu = 1/2$ ), in the single-pass (left) and multi-epoch (right) regimes.

datasets and three kernels in the *single-pass* (cfr. Section 1.3 and Table 4.2) schedule; the second, in Figures 4.9 and 4.10, puts the single-pass schedule next to the *multi-pass* (cfr. Section 1.3 and Table 4.2) one on one kernel per dataset.

The SQ and FQ curves are superposed at every value of  $M$  in both regimes. Considering the drafting procedure of the batch, a sample is revisited about once on average in the single-pass regime and about  $\sqrt{n}$  times in the multi-pass one, and FQ reuses its noise variable exactly as often. The two curves never differ by more than 0.01, and by less than 0.004 from  $M = 10^2$  on, against test errors of order 0.3 to 0.9. Empirically, the correlation introduced by freezing does not degrade the solution.

**Memory.** Addressing the low-memory setting considered at the beginning we recall that FQ forms the full-precision features once, quantizes them and keeps the one-bit

matrix; from then on it reads its features from store. SQ must form the full-precision value at every visit in order to draw a fresh noise variable, so it pays the evaluation exactly as FP does — it is the arithmetic that is low-precision, not the representation, as discussed in Section 1.3. The agreement between the two curves is what makes the frozen scheme usable in spite of not being covered.

At the largest grid point  $M = 10^4$  the frozen one-bit feature matrix of the training set occupies  $nM$  bits, about 10 MB, against the  $nM$  single-precision floats, about 320 MB, of its full-precision counterpart. Two costs should be kept apart here. Mini-batch SGD is chosen for the *optimization* cost: each step costs  $O(bM)$  and a pass over the data  $n/b$  steps, hence  $O(nM)$  per pass independently of  $b$ , and it regularizes implicitly through early stopping, whereas solving the normal equations in the  $M$ -dimensional feature space costs  $O(nM^2 + M^3)$  and requires an explicit penalty. That argument is independent of how the features are stored, and holds whether or not the matrix fits in memory — at  $M = 10^4$  the 320 MB matrix does fit comfortably, but its  $M \times M$  factorization is not cheap. The *storage* cost is a separate consideration. All four estimators must form the full-precision value  $\cos(\omega_j^\top x_i + \xi_j)$  at least once; what distinguishes them is how often. FQ and DQ form it when they build the matrix and discard it immediately afterwards, whereas FP and SQ form it at every visit of the sample. In particular SQ pays the recomputation without buying any storage saving, since a fresh dither is required at each step: the one-bit representation pays off only once it is frozen.

Conversely, having room for the  $nM$  matrix does not make SGD unnecessary, because the two budgets are distinct. Solving the normal equations in the feature space requires the  $M \times M$  covariance in working memory together with its factorization: at  $M = 10^4$  this is 400 MB and about  $10^{12}$  operations, against the  $b \times M$  numbers and the  $O(nM)$  operations per pass of the recursion. Storage and optimization are therefore relaxed by separate means, quantization by the former and mini-batch SGD by the latter, and the two are independent: one may hold the feature matrix and still be unable to factorize in the feature space.

**A consideration about kernel choice.** The comparison across the three kernels is a reminder that the choice of kernel dominates the effect we are measuring: on SUPERCONDUCT the Matérn kernel with  $\nu = 1/2$  reaches a test error about 10% below the Gaussian one. This does not make the Matérn kernel a default, nor DQ a better estimator than SQ. Both advantages come from the same source, an RKHS that contains functions of low smoothness, which suits these two targets; on the synthetic problem of Figure 4.4, where the target is smooth and lies in the Gaussian RKHS by construction, the ordering is reversed and DQ stays above SQ throughout the trajectory in both regimes. The kernel encodes an assumption on  $f_\rho$  and is selected on the problem; what our analysis says is that, once it is selected, one bit per coordinate costs nothing in rate or in features.

## Conclusion

**What the thesis does.** The starting point is the observation that random Fourier features remove the  $O(n^2)$  memory cost of kernel methods but replace it with the cost of storing an  $n \times M$  matrix of floating-point numbers, which is itself prohibitive in a memory-constrained system. We therefore studied least-squares regression solved by mini-batch SGD on random features that are stochastically rounded to one bit per coordinate, and asked what this costs statistically. Prior work only partially addresses this question: [20] analyze a kernel-matrix approximation under feature quantization, showing that rounding introduces an effective ridge term on  $\hat{K}$ , while [16] focus strictly on kernel approximation error for one-bit features using an asymmetric quantization scheme. Neither provides guarantees on the excess risk of the trained predictor.

**The theoretical contribution.** Chapter 3 propagates the quantization noise through the SGD recursion in the population  $L^2(\rho_X)$  norm. The rounding perturbs the covariance operator by a diagonal term  $\bar{D}$  of order  $1/M$ , producing a tracking bias between the quantized and the full-precision trajectories, bounded in Proposition 3.4.2, and an additional gradient variance, bounded in Proposition 3.4.6. Combining the two with the full-precision bound of [4] gives Theorem 3.3.1: the squared bias is of order  $\mu T/M^2$  and the extra variance of order  $\mu \log T$ , so that under the parameter choices of [4], for which  $\mu T \asymp \sqrt{n}$ , both are  $\tilde{\mathcal{O}}(1/\sqrt{n})$  as soon as  $M = \tilde{\mathcal{O}}(\sqrt{n})$ . Corollary 3.3.2 records the consequence: one-bit stochastic quantization preserves the minimax rate  $\mathcal{O}(1/\sqrt{n})$  of [3] with the same number of features as in full precision. No penalty is added anywhere: the regularization is implicit, and is carried by the step size, the horizon and the number of features. Hence, under the assumptions stated in Chapter 3, the statistical cost of low-precision arithmetic, performed via quantization, scales at the same rate as the full-precision setting. Moreover, given that the expected kernel in our analysis is the same as the initial one, no artifact over the function space is introduced to make up for the loss of precision.

**The empirical contribution.** Chapter 4 tests this on CASP and SUPERCONDUCT, with the Gaussian and the Matérn kernels, and separates two settings that the theory does not distinguish: the streaming estimator that the analysis describes, which redraws the noise variable at every visit and stores no representation, and the frozen estimator, which quantizes once and is the one that actually keeps one bit per coordinate. The two agree empirically, including in the multi-epoch schedule, where the frozen estimator reuses each noise variable  $\sqrt{n}$  times. Freezing does not destroy unbiasedness — the quantizer remains unbiased conditionally on the feature — but it destroys the independence between  $\gamma_i$  and  $w_i^q$  on which the analysis rests; the experiments suggest that this correlation is harmless in practice.

**A first step towards the deterministic estimator.** Unbiasedness of the quantizer is the core idea behind the proof, but the experiments show that it is not what governs accuracy. Deterministic sign quantization, which our framework does not cover, is comparable to full precision and frequently better, because it learns in the RKHS of  $\kappa_{q,q}^\Delta = \sum_{h \text{ odd}} |S_h|^2 \kappa^\Delta(h \cdot)$ , which contains  $\mathcal{H}_{\kappa^\Delta}$  since  $\kappa_{q,q}^\Delta - c \kappa^\Delta$  is positive definite for  $c \leq |S_1|^2 = 4/\pi^2$ . Given  $(\Omega, \xi)$  that map is non-random, so there is no rounding noise to propagate and (3.25) is the algorithm of [4] for the feature  $q(\omega^\top x + \xi)$ , with  $|q| \equiv 1$ ; what changes is the target, and a rate for it would be a statement about  $\mathcal{H}_{\kappa_{q,q}^\Delta}$  of the form:  $\|f_T - f_{\mathcal{H}_{q,q}}\|_\rho$ , where  $f_{\mathcal{H}_{q,q}}$  is the optimal estimator in the induced function space, which may be strictly bigger. This is why the present analysis has to come first: it shows that one bit quantization is coherent with the full-precision regime, so that the gap observed in Chapter 4 is attributable to the model choice and not to the bit depth. Our analysis explains why stochastic quantization is safe; it does not establish that it is the right choice.

### Open problems.

- The frozen estimator, which is the one that actually stores one bit per coordinate, is covered in no parameter configuration. Under the sampling scheme of [4], with replacement, a constant fraction of the samples is revisited, the noise terms are reused and  $\mathbb{E}[H_t | \mathcal{F}_t] = 0$  fails; under sampling without replacement the mini-batches are no longer independent and both the full-precision term of our decomposition and the independence arguments of Propositions 3.4.2 and 3.4.6 lose their justification. Closing this gap is a question left open here.
- Assumption (A6) is verified when the mini-batch operators are replaced by their limit in expectation, and assumed in general. A proof in the regime  $b < n$ , along the lines of [10], would remove it.
- Our bound holds in expectation over the mini-batches; a high-probability version would be more informative, and the rate itself is the one obtainable under no

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assumption on the spectrum of  $C_M$ . Refined source and capacity conditions, of the kind used in [3] and [4], would give faster rates and, presumably, a smaller number of features.

- Only one bit is analysed and tested. The analysis extends formally to  $b$ -bit rounding by replacing the bound  $2/M$  on the diagonal of  $D(x)$  with  $\delta_b^2/M$ ,  $\delta_b^2 = 2/(2^b - 1)^2$  [20], but the resulting trade-off between bits and features is not explored here. Neither is the deterministic estimator, which the experiments favour and which our framework does not cover.



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